

A Likelihood Bridge for Bayesian Meta-Analysis With Dependent Effect Sizes

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ABSTRACT. Primary studies in education and the behavioral sciences often report multiple, statistically dependent effect sizes, and the correlations needed to model that dependence are rarely reported. Frequentist meta-analysis handles this problem with robust variance estimation, which pairs a working model for estimation with a cluster-robust standard error that remains valid when the model is wrong. A Bayesian analyst who fits the same working model has no comparable option, because the posterior covariance reflects the prior and the assumed working correlation rather than the robust sandwich. A robust report therefore requires calibrating the posterior covariance to an external target without moving the point estimates, and the meta-analytic literature has never named that target. In this article, we show that Bayesian survey inference solved the same problem, and we prove the likelihood bridge that carries its solution into meta-analysis. Under matched conditions, a dependent-effects meta-analysis and a clustered Bayesian survey analysis are the same model under two interpretations, so the covariance to report is essentially the object the robust sandwich already targets. A center-preserving adjustment moves the posterior covariance onto this target while leaving every estimate unchanged, and we illustrate the framework on two published syntheses.

Keywords: dependent effect sizes; Bayesian meta-analysis; robust variance estimation; calibrated Bayes; marginal fixed-effect sandwich

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1. Introduction

In education and the behavioral sciences, primary studies rarely contribute a single effect size to a research synthesis. A study might report effects for several outcome measures, for multiple treatment contrasts that share a control group, for separate subgroups, or for repeated follow-up occasions. When these estimates address the review question, current guidance often favors retaining all of them rather than reducing each study to a single number (Page et al., 2015), and taxonomies of effect-size dependence describe how varied the resulting data structures can be (Ahn et al., 2012; López-López et al., 2018; Tipton et al., 2019). The data that reach the analysis stage are therefore clustered: each study contributes a vector of related effect sizes.

Modeling these data directly would require covariance information that primary studies rarely provide. Effect sizes estimated on the same participants, instruments, and treatment arms have sampling errors that covary, and a multivariate likelihood would encode this dependence through the within-study covariances, or equivalently through the correlations among the reported estimates. In practice, these correlations are seldom reported (Hong et al., 2018; Wei & Higgins, 2013). The analyst therefore knows that the estimates are dependent but usually cannot observe the dependence structure that an exact multivariate sampling model would require.

Robust variance estimation (RVE) changed the inferential problem by making the missing correlations far less consequential for fixed-effect inference. The analyst specifies a convenient covariance model for estimation, such as a correlated-effects or correlated-and-hierarchical-effects working model, and then reports a cluster-robust sandwich standard error for the fixed effects (Hedges et al., 2010; Pustejovsky & Tipton, 2022; Tipton & Pustejovsky, 2015). The working covariance organizes the fit, while the sandwich protects the reported uncertainty against misspecification of that working covariance, with small-sample refinements available for moderator tests at realistic numbers of studies (Tipton, 2015; Tipton & Pustejovsky, 2015). This separation between the model used for estimation and the covariance used for reporting is what made dependent-effects meta-regression routine.

The same separation does not come automatically in a Bayesian analysis. A Bayesian correlated-effects model fit at a fixed working correlation produces a posterior distribution for the fixed effects and heterogeneity components, and the covariance of that posterior reflects the prior curvature, the likelihood curvature, and the working covariance imposed by the fitted model rather than the cluster structure of the sampling errors. Treating posterior spread from the working model as if it were

robust uncertainty would therefore conflate the model one fits with the covariance one reports. A Bayesian analogue of robust variance estimation must instead keep the posterior center supplied by the working model and realign the reported covariance to an external target, in the calibrated-Bayes sense of using posterior inference while calibrating its frequency-relevant spread (Little, 2012; Rubin, 1984).

This narrows the problem considerably, because once the working model has been fit, the posterior mean and other point summaries are determined; the only open reporting decision is the covariance block attached to the moderator estimates or contrasts. The question is therefore which fixed-effect covariance target should replace the model-based posterior covariance when a robust report is desired. The meta-analytic literature supplies the working models and the frequentist sandwich vocabulary, but it has not named this target for a Bayesian fit.

A suitable target has already been developed in a neighboring literature. Bayesian survey inference separates the model used for estimation from the covariance needed for design-respecting inference in the same way: a working likelihood or pseudo-likelihood supplies the posterior center, and a design-based sandwich supplies the covariance to which that posterior should be calibrated. M. R. Williams and Savitsky (2021) resolve the gap by preserving the posterior center and recoloring its covariance to the sandwich target, and related pseudo-Bayesian mixed-model work supplies the clustered setting in which this principle is implemented (Savitsky & Toth, 2016; Savitsky & Williams, 2022). If a dependent-effects meta-analysis and a clustered Bayesian survey analysis share the same Gaussian estimating structure on the relevant boundary, then the target developed for surveys applies directly to meta-analysis.

In this article, we establish that bridge. We first show that, under a matched, unit-weighted specification, a correlated-effects meta-analysis and a unit-weighted clustered survey analysis evaluate the same conditional clusterwise Gaussian likelihood. Theorem 1 states this likelihood identity: with matched data and design matrices, a matched equicorrelated within-study covariance, and unit weights that are inert at both the observation level and the cluster (or random-effect) level, the two interpretations have identical cluster log likelihoods. The working correlation in this specification is fixed and conditional; it is not estimated, it is not integrated over in the posterior, and it is not the marginal intraclass correlation induced once the study-specific random effect is added.

We then separate the model one fits from the covariance one reports. The model is fit at the fixed working correlation, and its posterior mean remains the reported center; the covariance attached to that center is a separate reporting object. Theorem 2 identifies this object as the marginal fixed-effect sandwich, meaning the sandwich for the fixed effects and the between-study heterogeneity together, obtained by integrating over the study-specific random effects. The target is likelihood-only, so prior curvature does not enter it, and it carries the uncertainty in the heterogeneity into the reported covariance rather than treating the heterogeneity as known; the marginal likelihood retains a heterogeneity-dependent log-determinant term that is part of what makes the identity exact. Corollary 1 then supplies the reporting operation, an affine and center-preserving recoloring that maps the posterior covariance of a selected fixed-effect block or contrast onto the marginal-sandwich target while leaving the posterior mean unchanged.

The results are proved under explicit conditions, and it is useful to state at the outset what they do and do not deliver. The identity requires unit weights at both levels, a matched equicorrelated within-study covariance with a known per-cluster scale, matched data and design matrices, and a fixed conditional working correlation; it does not assert an unconditional equivalence between Bayesian meta-analysis and survey inference. Effect-specific heteroscedastic within-study variances and non-unit-weight informative designs require additional theory. The calibration is also not a finite-sample coverage theorem, because coverage depends on the interval or reference-distribution procedure layered on top of the covariance target. Small-sample devices such as CR2 corrections and Satterthwaite degrees of freedom are optional layers on the calibrated covariance, not alternative definitions of the target.

The bridge also clarifies how the present results relate to existing equivalence theory for dependent effect sizes. Pustejovsky and Chen (2024) relate meta-analytic representations through proportional-likelihood equivalency and a one-row-per-study aggregation route. That route is unavailable when a within-study moderator cannot be represented by study-level categorical subgrouping, as with a continuous effect-level moderator, although the unaggregated studies-as-clusters estimand remains defined in those cases. The marginal fixed-effect sandwich identified here reports uncertainty for that estimand, so the survey calibration target covers the cases that the aggregation route cannot.

The theory developed here also provides the foundation for an operational method. The present article identifies the covariance target and proves the bridge that defines

it, while a companion paper develops the resulting recoloring into a finite-sample procedure with small-sample corrections, an R implementation, and a simulation study of its coverage. The remainder of the article proceeds as follows. Section 2 reviews the two literatures that meet in the argument, dependent-effects robust variance estimation and multivariate meta-analysis on one side and clustered Bayesian survey inference and calibrated Bayes on the other, and fixes the notation. Section 3 defines the matched clustered Gaussian model the two traditions share and proves the likelihood identity together with its posterior consequences. Section 4 imports the survey calibration framework to define the covariance target for the reported fixed-effect block and derives the center-preserving recoloring that attains it. Section 5 illustrates the identity, the target, and the recoloring using two published syntheses. Section 6 discusses what carries across the bridge, what does not, and the questions the bridge opens.

2. Two Literatures for One Clustered Model

The covariance-target question involves two methodological literatures that built the same clustered-regression skeleton, with independent clusters, correlated within-cluster observations, fixed effects, and cluster-specific random effects, in largely separate vocabularies and with little mutual citation. In this section, we review what each literature supplies, explain why neither names the target on its own, and fix the notation in which the two are joined.

Multivariate models for dependent effect sizes entered meta-analysis early (Becker, 1992; Berkey et al., 1998; Kalaian & Raudenbush, 1996; Raudenbush et al., 1988; van Houwelingen et al., 2002) and were later extended through structural equation formulations (Cheung, 2008, 2013) and multilevel representations (Konstantopoulos, 2011; Van Den Noortgate et al., 2015). All of these models require the within-study covariance structure among effect-size estimates, and the correlations that structure needs are the ones primary studies seldom report (Hong et al., 2018; Wei & Higgins, 2013). Pseudo-likelihood and approximate-likelihood constructions relax the requirement by replacing the full multivariate likelihood with working components (Chen et al., 2015; Guolo, 2023; Hu et al., 2025), and composite-likelihood formulations have been carried into network settings (Liu et al., 2025). These approaches share a common feature: the fitted covariance model is acknowledged to be a working model, in the sense that it is chosen for tractability and is likely simpler than the true dependence structure.

Robust variance estimation made this approximation safe for fixed-effect inference. Hedges et al. (2010) paired an approximate working covariance with a cluster-robust sandwich estimator that remains valid for fixed-effect inference when the working covariance is wrong, small-sample corrections made the sandwich usable at realistic numbers of studies (Tipton, 2015; Tipton & Pustejovsky, 2015), and Pustejovsky and Tipton (2022) organized the practice around the correlated-effects and correlated-and-hierarchical-effects working models now standard in applied synthesis (Tanner-Smith et al., 2016). The same architecture, in which the estimating model is kept simple and a sandwich covariance respects the clustering, has counterparts in generalized estimating equations, where the working correlation is fixed by choice rather than estimated from scarce information (Liang & Zeger, 1986), and in the broader cluster-robust literature in statistics and econometrics (Bell & McCaffrey, 2002; Cameron & Miller, 2015; Imbens & Kolesár, 2016; MacKinnon et al., 2023; Pustejovsky & Tipton, 2018). In all of these traditions, the reported covariance is defined by an estimating-equation target, in the design-based sense of Binder (1983), and the model-based covariance is the object the sandwich replaces.

Bayesian meta-analysis has developed on the estimation side of this division but not on the reporting side. Prior specification for heterogeneity parameters has been examined closely (Röver et al., 2021; D. R. Williams et al., 2018), and Bayesian machinery for dependent and multivariate effect sizes remains an active area, from location-scale formulations (Jansen & Nestler, 2026) to robust model-averaged multilevel synthesis under publication bias (Bartoš et al., 2026). Recent work makes the practical gap explicit: Bartoš et al. (2026) consider cluster-robust standard errors for dependent effect sizes and set them aside because no Bayesian counterpart currently exists. The nearest construction is that of Wang et al. (2025), who calibrate a Bayesian composite-likelihood analysis of multivariate network meta-analysis to its Godambe sandwich through a post-sampling covariance adjustment. Their work confirms that sandwich-calibrated Bayesian synthesis is feasible. However, it targets the composite likelihood of a network structure rather than the correlated-effects working model of dependent-effects meta-regression, and it does not connect the calibration to the robust variance estimation tradition that governs how dependent effect sizes are reported in education and the behavioral sciences. The specific combination needed there, a named covariance target for the Bayesian correlated-effects fit that is reached without moving the posterior center, has remained open.

The missing reporting step, however, has already been developed in survey statistics. Design-based inference defines parameters through population estimating equations and reports the corresponding cluster sandwich (Binder, 1983), and survey-weighted pseudo-likelihoods carry the same logic into model-based estimation under complex sampling (Pfeffermann, 1993; Pfeffermann et al., 1998; Rabe-Hesketh & Skrondal, 2006). On the Bayesian side, pseudo-posterior constructions establish consistency under informative sampling (Savitsky & Toth, 2016), with extensions to mixed models and cluster sampling (León-Novelo & Savitsky, 2023; Savitsky & Williams, 2022). M. R. Williams and Savitsky (2021) complete the reporting step. Because the pseudo-posterior covariance does not match the design-based sandwich, they calibrate the posterior, preserving its center while realigning its covariance to the sandwich target; this is an instance of the calibrated-Bayes principle articulated by Rubin (1984) and Little (2012). It is important to note that this literature concerns surveys and never mentions meta-analysis. Carrying its calibration target across to the Bayesian correlated-effects meta-analysis is the step taken in the present article, and Section 3 proves the bridge that licenses the transfer.

Within meta-analysis, the closest structural precedent for such a bridge is the equivalence family of Pustejovsky and Chen (2024), who relate ad hoc data strategies to multivariate models through proportional likelihoods and record that the corresponding Bayesian posteriors coincide under matched priors. Their equivalences run through a one-row-per-study aggregation route, which exists when moderators are constant within study or can be represented by study-level categorical subgrouping. When a moderator varies within study in a way that subgrouping cannot express, for example a continuous effect-level moderator, the aggregated representation has no left-hand side, while the unaggregated studies-as-clusters model remains fully defined. The present argument therefore works directly with the unaggregated model, and it adds a second interpretation from outside meta-analysis. The survey interpretation contributes the covariance target that the meta-analytic literature left unnamed.

Notation. The notation is fixed here and used throughout. Clusters are indexed $j = 1, \dots, J$, and units within cluster are indexed $i = 1, \dots, n_j$. For each cluster, $\mathbf{y}_j$ is the response vector and $\mathbf{X}_j$ is the $n_j \times p$ observation-level design matrix, whose rows may vary within cluster; this property matters throughout, because it is what lets effect-level moderators enter the model without aggregation. The fixed-effect vector is $\boldsymbol{\beta} \in \mathbb{R}^p$, cluster random effects are θ_j , and the between-cluster variance is τ^2 . The within-cluster working covariance is $\mathbf{V}_j = a_j \mathbf{C}_j(\rho)$, where $a_j > 0$ is a known

per-cluster scale, $\mathbf{C}_j(\rho)$ is the equicorrelation matrix defined in Section 3, and ρ is a fixed conditional residual working correlation, that is, the within-cluster correlation of $\mathbf{y}_j \mid (\boldsymbol{\beta}, \theta_j)$; it is not the marginal intraclass correlation, which the random intercept inflates. The survey interpretation carries an observation weight $\tilde{w}_{ij}$ and a cluster (random-effect) weight $\tilde{w}_j$, and throughout the paper both are set to one, the unit-weight boundary. Three parameter vectors then organize the argument:

$$\boldsymbol{\eta} = (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top)^\top, \quad \boldsymbol{\zeta} = (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top, \tau^2)^\top, \quad \boldsymbol{\xi} = (\boldsymbol{\beta}^\top, \tau^2)^\top. \quad (1)$$

Each vector serves a distinct purpose. The vector $\boldsymbol{\eta} = (\boldsymbol{\beta}, \boldsymbol{\theta})$ indexes the conditional joint parameterization and carries the obstruction to full-joint calibration, because Lemma 1 shows that its sandwich is rank-deficient. The vector $\boldsymbol{\zeta} = (\boldsymbol{\beta}, \boldsymbol{\theta}, \tau^2)$ indexes posterior probability-measure statements, because τ^2 belongs to the hierarchical posterior. The vector $\boldsymbol{\xi} = (\boldsymbol{\beta}, \tau^2)$ indexes the marginal estimating problem whose $\boldsymbol{\beta}$ -block the analyst reports, and Theorem 2 transports the marginal sandwich in $\boldsymbol{\xi}$ across the two interpretations. The study $\leftrightarrow$ cluster and effect-size $\leftrightarrow$ observation crosswalk appears in Figure 1; the complete symbol crosswalk is recorded in Appendix A.

3. The Shared Clustered Gaussian Model and the Likelihood Identity

In this section, we define the clustered Gaussian model that the two traditions share and prove that they assign it the same clusterwise likelihood. Every later result, from the covariance target of Section 4 to the recoloring computed in Section 5, is defined on this single model.

For each cluster $j = 1, \dots, J$, let the response vector be $\mathbf{y}_j = (y_{1j}, \dots, y_{n_jj})^\top$ and the design matrix be $\mathbf{X}_j \in \mathbb{R}^{n_j \times p}$. The design matrix is observation-level, meaning its rows may differ within a cluster, so effect-level moderators in the meta-analytic interpretation and observation-level covariates in the survey interpretation enter directly, never as cluster averages. With fixed effects $\boldsymbol{\beta} \in \mathbb{R}^p$ and a latent cluster effect θ_j , the shared conditional model is

$$\mathbf{y}_j \mid \boldsymbol{\beta}, \theta_j \sim \mathbf{N}_{n_j}(\mathbf{X}_j \boldsymbol{\beta} + \theta_j \mathbf{1}_{n_j}, \mathbf{V}_j), \quad (2)$$

$$\theta_j \mid \tau^2 \sim \mathbf{N}(0, \tau^2), \quad j = 1, \dots, J. \quad (3)$$

In the meta-analytic interpretation, θ_j is the study-specific deviation and τ^2 is the between-study heterogeneity; in the survey interpretation, θ_j is the cluster random effect and τ^2 is the between-cluster variance. The model equations are identical under the two interpretations, and only the vocabulary attached to them differs. Figure 1

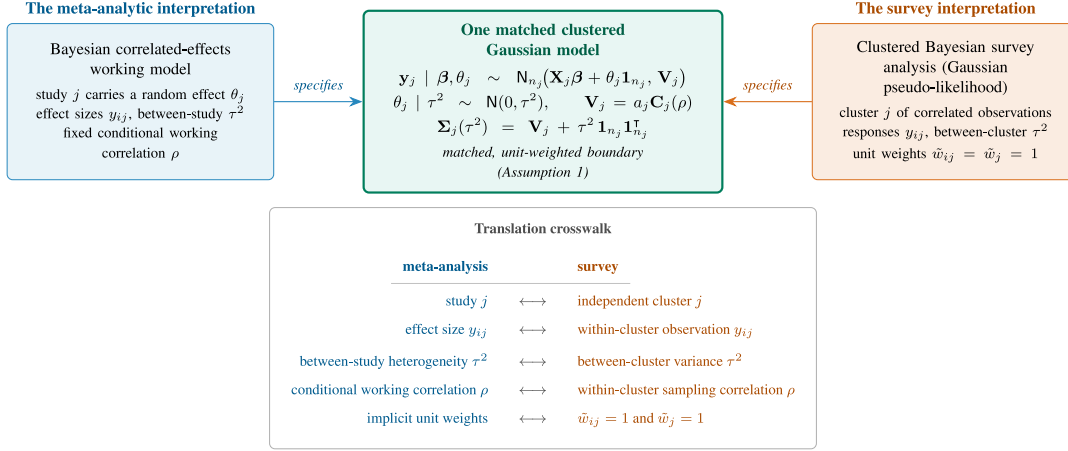


FIGURE 1. Two interpretations of one clustered Gaussian model. The correlated-effects meta-analysis (left) and the unit-weighted clustered Bayesian survey analysis (right) specify the same matched clustered Gaussian model (center), with the same fixed conditional working correlation ρ and unit weights at both the observation and the cluster level. The crosswalk table records the translation between the two vocabularies; Theorem 1 states the resulting likelihood identity under Assumption 1.

displays the full crosswalk: study $\leftrightarrow$ cluster, effect size $\leftrightarrow$ within-cluster observation, one matched equicorrelated (compound-symmetric) covariance, and unit weights at both levels.

The working covariance takes the equicorrelated form

$$\mathbf{C}_j(\rho) = (1 - \rho)\mathbf{I}_{n_j} + \rho \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top, \quad (4)$$

$$\mathbf{V}_j = a_j \mathbf{C}_j(\rho), \quad a_j > 0, \quad (5)$$

which is positive definite for $-\frac{1}{n_j-1} < \rho < 1$ when $n_j > 1$. The working correlation ρ is fixed in advance and matched across the two interpretations; it is not estimated and not integrated over in the posterior, in the same spirit in which generalized estimating equations fix a working correlation by choice (Liang & Zeger, 1986). The scale $a_j > 0$ is known, common to both interpretations at each j , and free to vary across clusters. Note that ρ parameterizes the conditional model $\mathbf{y}_j \mid \boldsymbol{\beta}, \theta_j$: it is a conditional residual working correlation, not the marginal intraclass correlation, which the random intercept θ_j inflates through the marginal covariance defined next. Equicorrelation, the fixed ρ , and the known per-cluster scales are working-model and reporting-scope choices; the exactness of the results that follow comes from matching

the known within-cluster covariance $\mathbf{V}_j$ across interpretations, not from the particular form chosen for it.

Integrating θ_j out of the conditional model gives the marginal covariance and the marginal model,

$$\Sigma_j(\tau^2) = \mathbf{V}_j + \tau^2 \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top, \quad (6)$$

$$\mathbf{y}_j \mid \boldsymbol{\beta}, \tau^2 \sim \mathbf{N}_{n_j}(\mathbf{X}_j \boldsymbol{\beta}, \Sigma_j(\tau^2)), \quad (7)$$

whose log-determinant term $\log |\Sigma_j(\tau^2)|$ depends on τ^2 and therefore cannot be dropped. This term matters later, because it is part of the parameter-dependent geometry of the marginal likelihood on which Section 4 builds.

The survey interpretation enters through the weighted pseudo-likelihood of the survey-calibration tradition (Savitsky & Toth, 2016; Savitsky & Williams, 2022). For cluster j , with observation weights $\tilde{w}_{ij}$ and a cluster (random-effect) weight $\tilde{w}_j$, the clusterwise pseudo-likelihood contribution is

$$L_j^{\text{pseudo}}(\boldsymbol{\beta}, \theta_j, \tau^2) = p(\mathbf{y}_j \mid \boldsymbol{\beta}, \theta_j; \mathbf{V}_j)^{\tilde{w}_j} \cdot p(\theta_j \mid \tau^2)^{\tilde{w}_j}, \quad (8)$$

where $p(\mathbf{y}_j \mid \boldsymbol{\beta}, \theta_j; \mathbf{V}_j)$ is the clusterwise multivariate Gaussian density with the observation weights $\tilde{w}_{ij} = 1$ inert inside it. Setting $\tilde{w}_j = 1$ collapses (8) to $p(\mathbf{y}_j \mid \boldsymbol{\beta}, \theta_j; \mathbf{V}_j) p(\theta_j \mid \tau^2)$, the unweighted clusterwise Gaussian likelihood of the meta-analytic interpretation. When $\rho = 0$ the clusterwise density factors into a product of univariate densities, recovering the familiar independence form; for $\rho \neq 0$ the covariance $\mathbf{V}_j$ has off-diagonal entries and the multivariate form must be retained. This correlated-cluster Gaussian pseudo-likelihood is the present paper's own construction, motivated by the survey calibration program of Savitsky and Williams (2022) rather than drawn from it; the survey literature supplies the calibration principle, and the object it acts on here is built to match the meta-analytic model exactly.

Definition 1 (Common clustered Gaussian object). Independent clusters $j = 1, \dots, J$; ordered response vectors $\mathbf{y}_j$; observation-level design matrices $\mathbf{X}_j$; fixed effects $\boldsymbol{\beta}$; latent effects $\theta_j \mid \tau^2 \sim \mathbf{N}(0, \tau^2)$; known positive-definite $\mathbf{V}_j = a_j \mathbf{C}_j(\rho)$ with $a_j > 0$; conditional model (2); marginal covariance (6); parameter vectors $\boldsymbol{\eta} = (\boldsymbol{\beta}, \boldsymbol{\theta})$, $\boldsymbol{\zeta} = (\boldsymbol{\beta}, \boldsymbol{\theta}, \tau^2)$, $\boldsymbol{\xi} = (\boldsymbol{\beta}, \tau^2)$.

Assumption 1 (Matched unit-weighted clustered Gaussian boundary). The meta-analytic and survey interpretations share: (A1) the same cluster partition; (A2) the same ordered response vector $\mathbf{y}_j$; (A3) the same observation-level design matrix

$\mathbf{X}_j$ (rows may vary within cluster); (A4) a fixed matched working correlation ρ (conditional residual, not estimated or posterior-integrated); (A5) the same known positive-definite covariance $\mathbf{V}_j = a_j \mathbf{C}_j(\rho)$; (A6) a common known per-cluster scale $a_j > 0$ (a per-cluster match, free to vary with j); (A7) unit observation weight $\tilde{w}_{ij} = 1$ and unit cluster (random-effect) weight $\tilde{w}_j = 1$ —both levels equal to one, the unit-weight boundary at which the weights in (8) are inert. Both interpretations use a Gaussian kernel, and matched priors are imposed wherever posterior claims are made.

The restricted translation map $\mathcal{T}_A$ identifies the two interpretations only under Assumption 1. Under changed designs, an estimated ρ , a scale mismatch, non-unit weights, or non-matched covariances, the identification fails, and the domain ledger in Appendix A records how. Two tiers of generality are worth noting. Theorem 1 below is stated for the matched equicorrelation family $\mathbf{V}_j = a_j \mathbf{C}_j(\rho)$ with arbitrary per-cluster scales, while Theorem 2 requires only a matched marginal Gaussian kernel. Cluster-constant designs (a condition we label S1, examined in Appendix B) simplify some expressions but are not required anywhere; effect-specific within-cluster heteroscedasticity lies outside Assumption 1 and is taken up in Section 6.

Under the conditional parameterization $\boldsymbol{\eta} = (\boldsymbol{\beta}, \boldsymbol{\theta})$, the clusterwise Gaussian log likelihood is

$$\ell_j(\boldsymbol{\eta}) = -\frac{1}{2} \left[n_j \log(2\pi) + \log |\mathbf{V}_j| + \mathbf{r}_j(\boldsymbol{\eta})^\top \mathbf{V}_j^{-1} \mathbf{r}_j(\boldsymbol{\eta}) \right], \quad (9)$$

with cluster residual $\mathbf{r}_j(\boldsymbol{\eta}) = \mathbf{y}_j - \mathbf{X}_j \boldsymbol{\beta} - \boldsymbol{\theta}_j \mathbf{1}_{n_j}$. Let $\ell_j^M(\boldsymbol{\eta})$ denote the meta-analytic interpretation of this function and $\ell_j^S(\boldsymbol{\eta})$ the survey interpretation at the unit-weight boundary, with totals $\ell^R(\boldsymbol{\eta}) = \sum_{j=1}^J \ell_j^R(\boldsymbol{\eta})$ for $R \in \{M, S\}$.

Theorem 1 (Clusterwise likelihood identity). Let Assumption 1 hold and let the two interpretations be identified through the restricted translation map $\mathcal{T}_A$. Under the matched multivariate-normal normalization,

$$\ell_j^M(\boldsymbol{\eta}) = \ell_j^S(\boldsymbol{\eta}) \quad \text{for every cluster } j \text{ and every } \boldsymbol{\eta}, \quad (10)$$

and therefore

$$\ell^M(\boldsymbol{\eta}) = \ell^S(\boldsymbol{\eta}). \quad (11)$$

The equality is exact: under the matched normalization the additive constant is zero. Under an alternative admissible normalization the two log likelihoods may differ, but only by a cluster-additive term free of every parameter being optimized, differentiated, or integrated. We provide a proof in Appendix B.

The implication of Theorem 1 is that the correlated-effects meta-analysis and the unit-weighted clustered survey analysis share a single likelihood, written in two vocabularies. Every quantity computed from that likelihood, including every estimate, score, and curvature, is therefore the same object under both interpretations, with no additive constant to reconcile. Where Pustejovsky and Chen (2024) establish proportionality between meta-analytic representations, the matched normalization here yields equality with a zero constant, which is what allows derivative-based objects to transfer without further bookkeeping.

Two boundary features of $\mathcal{T}_A$ deserve formal notice, because each marks a condition under which the identity fails. **Caveat 1 (Scale mismatch is parameter-dependent).** The per-cluster scale match in Assumption 1(A6) is essential: if the two interpretations carry different cluster scales, the log likelihoods differ by a parameter-dependent quadratic rather than a constant, so scores and curvature no longer agree. **Caveat 2 (The marginal determinant survives only under a matched covariance).** The term $\log |\Sigma_j(\tau^2)|$ in the marginal model (7) is parameter-dependent geometry and cannot be discarded; it agrees across interpretations only because Assumption 1(A5–A6) forces $\Sigma_j^M(\tau^2) = \Sigma_j^S(\tau^2)$ pointwise in τ^2 .

The identity does not require cluster-constant predictors. Assumption 1(A3) demands only that both interpretations use the same observation-level $\mathbf{X}_j$, whose rows may vary freely within cluster, and nothing in the construction ever averages a study down to one row. The identity therefore covers the syntheses in which moderators vary from one effect size to the next within a study. For such designs, literal one-row-per-study aggregation cannot represent the within-study contrast; where the moderator also resists study-level categorical subgrouping, as a continuous effect-level moderator does, the unaggregated studies-as-clusters estimand is the only one available.

Proposition 2 (Immediate consequences of the identity). Under the hypotheses of Theorem 1, the meta-analytic and survey likelihoods are the same function of $\boldsymbol{\eta}$, so their scores, Hessians, and cluster scores coincide: for every $\boldsymbol{\eta}$, $\nabla_{\boldsymbol{\eta}} \ell^M(\boldsymbol{\eta}) = \nabla_{\boldsymbol{\eta}} \ell^S(\boldsymbol{\eta})$ and $\nabla_{\boldsymbol{\eta}}^2 \ell^M(\boldsymbol{\eta}) = \nabla_{\boldsymbol{\eta}}^2 \ell^S(\boldsymbol{\eta})$. Under the same prior probability measure on $\boldsymbol{\zeta} = (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top, \tau^2)^\top$ (the same hyperparameters and the same $p(\boldsymbol{\theta} \mid \tau^2)$), with propriety and a finite posterior first moment, the two posterior probability measures are equal:

$$\Pi_M(A \mid \mathbf{y}) = \Pi_S(A \mid \mathbf{y}) \quad \text{for every measurable set } A \text{ in the shared parameter space.} \quad (12)$$

We provide a proof in Appendix B.

The implication of Proposition 2 is that the two interpretations share one posterior distribution and, with it, one center, the posterior mean. Posterior-measure equality is a property of the distribution itself, so it does not depend on the sampler or the software used to explore it. This shared center is the quantity that the calibration of Section 4 holds fixed while the reported covariance is adjusted. One scope clarification is needed for the later development. The score and Hessian equalities are statements in $\boldsymbol{\eta} = (\boldsymbol{\beta}, \boldsymbol{\theta})$, because τ^2 does not enter the conditional likelihood, while the posterior-measure equality (12) is stated in the full hierarchical vector $\boldsymbol{\zeta}$. A sandwich assembled in $\boldsymbol{\eta}$ is therefore a conditional latent-effect object, and Section 4 shows that this distinction is one reason the reported covariance must be marginal.

Under Assumption 1 and the translation map $\mathcal{T}_A$, the two interpretations therefore describe one matched clustered Gaussian likelihood, with one posterior measure and one shared center. Section 4 applies the survey calibration framework to this object.

4. Setting the RVE Covariance Target by Importing Survey Calibration

An analyst reports a moderator coefficient or a contrast from a fitted synthesis, that is, a selected block $\boldsymbol{\psi} = \mathbf{R}_\psi \boldsymbol{\beta}$ of the fixed effects. Section 3 established that the fitted model is one likelihood under two interpretations. In this section, we use the survey interpretation to answer the question posed in Section 1, namely which covariance the reported block should carry. We proceed in three steps: we first identify the estimand that the reported block estimates, we then show why a sandwich cannot be assembled in the full conditional parameterization, and we finally define the marginal sandwich target together with the recoloring that attains it.

Following the design-based estimating-equation framework of Binder (1983), the reported fixed-effect block estimates the pseudo-true value $\boldsymbol{\beta}^\dagger$, defined as the solution of the population-level marginal cluster-score equation under independent-cluster (studies-as-clusters) sampling. Because the working covariance $\mathbf{V}_j$ is an acknowledged approximation, the covariance attached to the estimate of $\boldsymbol{\beta}^\dagger$ should be that of the corresponding cluster M-estimator, which is the Godambe sandwich of the composite-likelihood tradition (Varin, 2008; Varin et al., 2011), rather than the curvature of the fitted model. The Bayesian fit supplies a different matrix: the posterior covariance $\text{Var}_\Pi(\boldsymbol{\psi} \mid \mathbf{y})$ carries prior curvature and inherits the working correlation built into the likelihood. The task is therefore to name the sandwich the reported block should match, and to move the posterior covariance onto it without disturbing the posterior mean.

One route can be ruled out at the start. Consider the conditional parameterization $\boldsymbol{\eta} = (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top)^\top$ with likelihood-only cluster scores $\mathbf{s}_j(\hat{\boldsymbol{\eta}}) = \nabla_{\boldsymbol{\eta}} \ell_j(\hat{\boldsymbol{\eta}})$ evaluated at a common point, and the conditional meat

$$\mathbf{M}_\eta = \sum_{j=1}^J \mathbf{s}_j(\hat{\boldsymbol{\eta}}) \mathbf{s}_j(\hat{\boldsymbol{\eta}})^\top. \quad (13)$$

Lemma 1 (Rank limitation of the conditional meat). $\text{rank}(\mathbf{M}_\eta) \leq J < p + J = \dim(\boldsymbol{\eta})$ whenever $p \geq 1$, so the conditional full-joint sandwich is singular. We provide a proof in Appendix C.

The implication of Lemma 1 is that full-joint calibration in $\boldsymbol{\eta}$ is not available: the meat is a sum of J rank-one cluster terms, one per latent effect, so its rank cannot reach the conditional dimension $p + J$ when $p \geq 1$. This is the incidental-parameter problem of Neyman and Scott (1948) appearing in the calibration setting (see also Lancaster, 2000), and a singular meat admits neither the inversion nor the matrix square root that a recoloring map requires. Lemma 1 does not disqualify the conditional $\boldsymbol{\beta}$ -block, which may well be positive definite. The reported covariance must be marginal for a more basic reason, namely the estimand itself: the reported quantity integrates over the study-specific effects $\boldsymbol{\theta}$, as the shared posterior center of Proposition 2 already does, so the covariance attached to it must be assembled in the marginal model, where τ^2 is carried as an explicit nuisance coordinate.

The marginal objects are assembled from the marginal model (7) of Section 3. Write $\ell^{\text{marg}}(\boldsymbol{\xi}) = \sum_{j=1}^J \ell_j^{\text{marg}}(\boldsymbol{\xi})$ for the likelihood-only marginal log likelihood in $\boldsymbol{\xi} = (\boldsymbol{\beta}^\top, \tau^2)^\top$, with cluster scores $\mathbf{s}_j^{\text{marg}}(\boldsymbol{\xi}) = \nabla_{\boldsymbol{\xi}} \ell_j^{\text{marg}}(\boldsymbol{\xi})$, and let $\hat{\boldsymbol{\xi}}$ be an interior root of the marginal cluster-score equation $\sum_j \mathbf{s}_j^{\text{marg}}(\boldsymbol{\xi}) = \mathbf{0}$ at which the bread $\mathbf{H}_\xi = -\nabla_{\boldsymbol{\xi}}^2 \ell^{\text{marg}}(\hat{\boldsymbol{\xi}})$ is positive definite. The marginal meat and sandwich are

$$\mathbf{M}_\xi = \sum_{j=1}^J \mathbf{s}_j^{\text{marg}}(\hat{\boldsymbol{\xi}}) \mathbf{s}_j^{\text{marg}}(\hat{\boldsymbol{\xi}})^\top, \quad (14)$$

$$\mathbf{V}_\xi^{\text{marg}} = \mathbf{H}_\xi^{-1} \mathbf{M}_\xi \mathbf{H}_\xi^{-1}. \quad (15)$$

Because $\mathbf{M}_\xi$ is a $(p+1) \times (p+1)$ matrix, the automatic rank deficiency of Lemma 1 no longer applies; full-rank use still requires the usual score-variation and estimability conditions, and at an exact marginal maximizer the cluster scores sum to zero, so the outer-product meat has rank at most $J - 1$. By partitioned inversion, the $\boldsymbol{\beta}$ -block bread of (15) is the Schur complement $\mathbf{S} = \mathbf{H}_{\beta\beta} - \mathbf{H}_{\beta\tau^2} \mathbf{H}_{\tau^2\tau^2}^{-1} \mathbf{H}_{\beta\tau^2}^\top$, and this is how uncertainty about the heterogeneity parameter propagates into the reported

block: the marginal target is a joint object in $(\boldsymbol{\beta}, \tau^2)$ rather than a $\boldsymbol{\beta}$ -only sandwich computed at a fixed τ^2 , and the parameter-dependent log-determinant of Caveat 2 is the term through which τ^2 retains its influence. For the reported block $\boldsymbol{\psi} = \mathbf{R}_\psi \boldsymbol{\beta}$, the selected target is

$$\mathbf{V}_\psi^{\text{target}} = \mathbf{R}_\psi \mathbf{V}_\xi^{\text{marg}}[\boldsymbol{\beta}, \boldsymbol{\beta}] \mathbf{R}_\psi^\top. \quad (16)$$

Definition 2 (Reported covariance target). The covariance target for the reported block $\boldsymbol{\psi} = \mathbf{R}_\psi \boldsymbol{\beta}$ is the selected $\boldsymbol{\beta}$ -block (16) of the full- $\boldsymbol{\xi}$ marginal sandwich (15): likelihood-only bread and meat, with prior curvature entering neither; Schur-complement $\boldsymbol{\beta}$ -block bread $\mathbf{S}$ propagating τ^2 uncertainty; evaluated at the interior score root $\hat{\boldsymbol{\xi}}$. The reported center is the posterior mean $\bar{\boldsymbol{\xi}}_{\text{post}}$, which the recoloring of Corollary 1 holds fixed. Three points carry distinct symbols: the population pseudo-true root $\boldsymbol{\xi}^\dagger$, the sample score root $\hat{\boldsymbol{\xi}}$ at which the sandwich is evaluated, and the posterior mean $\bar{\boldsymbol{\xi}}_{\text{post}}$ that centers the report; under the regularity conditions of Appendix C, both $\hat{\boldsymbol{\xi}}$ and $\bar{\boldsymbol{\xi}}_{\text{post}}$ are consistent for $\boldsymbol{\xi}^\dagger$. The target is an asymptotic, independent-cluster (CR0) object.

Definition 2 fixes the theorem-level target; in practice, optional layers are added on top of it. It is important to keep these layers distinct, because two confusions are common. The CR2 small-sample correction is a bias-reduced finite-sample estimator of the same sandwich, and Satterthwaite degrees of freedom are a reference-distribution adjustment; each may be applied to the calibrated block, and the companion paper develops and evaluates that finite-sample layer. Neither correction defines the target. Likewise, the profile block, meaning the $\mathbf{H}_{\beta\beta}$ -bread standard error computed with τ^2 held fixed, which is what standard cluster-robust meta-analysis software returns (Pustejovsky, 2026), is a practical estimator of the target; it is exact under Cox–Reid orthogonality ($\mathbf{H}_{\beta\tau^2} = \mathbf{0}$) and an approximation otherwise, with the gap governed by the cross-information block that the Schur bread includes and the profile bread omits.

As defined, the target is a survey-side object; it is the covariance a design-based survey statistician would calibrate toward for a cluster sample with unit weights. The following theorem, the marginal counterpart of Theorem 1, shows that it is equally a meta-analytic object.

Theorem 2 (Marginal fixed-effect sandwich identity). Under Assumption 1 and the matched translation $\mathcal{T}_A$, with the same random-effect distribution, observation-level design matrix, evaluation point, and bread–meat convention applied

on both interpretations, the marginal fixed-effect sandwiches are equal:

$$\mathbf{V}_{\xi, \mathbf{M}}^{\text{marg}} = \mathbf{V}_{\xi, \mathbf{S}}^{\text{marg}}. \quad (17)$$

We provide a proof in Appendix C.

The implication of Theorem 2 is that the covariance target a survey statistician would calibrate toward, following the center-preserving principle of M. R. Williams and Savitsky (2021), is algebraically the marginal fixed-effect object that the Bayesian correlated-effects meta-analysis already defines. The meta-analyst can therefore adopt the target without changing the unit of analysis or routing the synthesis through survey software, and the posterior center does not move. Because the proof relies only on the equality of matched marginal Gaussian densities, the same reasoning extends to any matched marginal covariance, of which the equicorrelated family (6) is the case the working models use.

The remaining step is the operation that moves the posterior covariance onto the target. Let $\bar{\boldsymbol{\psi}}_{\text{post}} = \mathbf{E}_{\Pi}(\boldsymbol{\psi} \mid \mathbf{y})$ denote the posterior mean of the reported block and $\boldsymbol{\Sigma}_{\text{post}, \psi} = \text{Var}_{\Pi}(\boldsymbol{\psi} \mid \mathbf{y})$ its posterior covariance; as with $\bar{\boldsymbol{\xi}}_{\text{post}}$, the bar marks a posterior functional, and hats remain reserved for score roots.

Corollary 1 (Center-preserving recoloring). When $\boldsymbol{\Sigma}_{\text{post}, \psi}$ and $\mathbf{V}_{\psi}^{\text{target}}$ are both positive definite, with Cholesky factors $\mathbf{L}_{\text{post}}$ and $\mathbf{L}_{\text{target}}$, the affine map $\mathcal{C}_{\psi}(\boldsymbol{\psi}) = \bar{\boldsymbol{\psi}}_{\text{post}} + \mathbf{L}_{\text{target}} \mathbf{L}_{\text{post}}^{-1}(\boldsymbol{\psi} - \bar{\boldsymbol{\psi}}_{\text{post}})$ fixes the center $\bar{\boldsymbol{\psi}}_{\text{post}}$ and maps the covariance of the selected block to $\mathbf{V}_{\psi}^{\text{target}}$.

We call this operation the center-preserving covariance calibration (CCC). Three attributions should be kept distinct. The calibration principle is that of M. R. Williams and Savitsky (2021), carried across the bridge; the finite-sample algebraic exactness of the map, meaning that the recolored covariance equals the target exactly rather than asymptotically, is the present paper’s contribution; and the target itself specializes the estimating-equation covariance of Binder (1983) to the independent-cluster design. Attribution notes recording what is imported and what is new appear in Appendix A.

In practice, the calibration is carried out in one pass. The analyst fits the Bayesian correlated-effects model at the fixed working correlation and retains the posterior mean and posterior covariance of the reported block; computes the marginal sandwich (15) at the score root with Schur bread and selects the block (16); checks that both covariance blocks are positive definite, as the recoloring requires (when $\hat{\tau}^2 = 0$ lies on

the boundary, either an interior-root scoping or a profile-bread fallback convention applies; the interior-root requirement and the positive-definiteness checks are recorded in Appendix C); applies the recoloring of Corollary 1; and, if a small-sample report is wanted, adds the CR2 and Satterthwaite layers on top of the calibrated block. The `bayesRVE` package automates each step.

Two diagnostics tell the analyst how much the calibration matters in a given synthesis: the percentage gap between the profile and full- ξ standard errors, and the size of the cross-information block, summarized by $\max |\mathbf{H}_{\beta\tau^2}|$ together with its scale-free companion $\|\mathbf{H}_{\beta\beta}^{-1/2}\mathbf{H}_{\beta\tau^2}\mathbf{H}_{\tau^2\tau^2}^{-1/2}\|_2$. Near Cox–Reid orthogonality both are negligible and the profile standard error already coincides with the target; where either is material, the analyst reports the recolored full- ξ block. The diagnostics indicate when recoloring matters; they do not define the target, and no universal threshold is implied.

We close this section with two boundary notes. First, the bridge, the target, and the recoloring hold on the matched unit-weighted clustered Gaussian boundary of Assumption 1, and the covariance statements concern which matrix is reported; the results make no finite-sample frequentist coverage claim, because coverage depends on the confidence-region method layered on top of the covariance. Second, two supporting results, the conditional latent-effect sandwich equality (Proposition 3) and the aligned HC/CR small-sample corrections (Corollary 2), transport only as matched-convention objects, with matched data, designs, working correlation, and covariances on both sides; they are stated and proved in Appendix D.

5. Illustration on Two Real Syntheses

In this section, we illustrate the three layers of the framework on real data: the likelihood identity of Theorem 1, the marginal target of Definition 2 with its diagnostics, and the recoloring of Corollary 1. We chose the two syntheses because they fall on opposite sides of the estimand-existence distinction discussed in Section 2. Both are fit at the unit-weighted $\rho = 0$ study-mean working boundary (the common per-cluster scale is the study mean of the sampling variances), and $\rho = 0.8$ is carried alongside as a labeled comparator.

The first synthesis is the reading-performance meta-analysis of Clinton-Lisell (2021), in which $J = 20$ studies contribute $K = 31$ effect sizes and the two moderators (pace and medium) are constant within every study, so within-study moderator variation is absent in 20 of 20 studies. For present purposes this design is degenerate,

and we label it Case B: one-row-per-study aggregation loses nothing, and the Pustejovsky and Chen (2024) route runs in parallel. The second is the enactment synthesis of Roberts et al. (2022), in which $J = 145$ studies contribute $K = 443$ effect sizes and a three-level task moderator varies within study in 28 of 145 studies across 118 effect sizes. This design is Case A for literal study-level collapse, because replacing each of those studies by one row would average the task indicators and lose the cell-mean contrast. Because the task moderator is categorical, a subgrouping construction could still represent it; the point of the example is that the studies-as-clusters target is computed directly on the unaggregated estimand, and the general case it stands for, within-study moderators that resist subgrouping such as continuous effect-level moderators, has no aggregated representation at all. We therefore compute the target in one synthesis where aggregation would also have worked and in one where literal aggregation would not.

The identity layer returns exact-zero residuals: for each synthesis and at each fixed $\rho \in \{0, 0.8\}$, the marginal log likelihood $\ell^{\text{marg}}(\tau^2)$, its analytic score, the observed information, and the assembled $\Sigma_j(\tau^2)$ were computed once under the meta-analytic interpretation and once under the survey interpretation. The two agree to $\max |\Delta| = 0$, exact floating-point equality, because under Assumption 1 both interpretations evaluate the same marginal Gaussian likelihood (7). Figure 2 displays the overlay at $\rho = 0$. Within-study moderator variation does not disturb the identity in either synthesis, as Proposition 1 of Appendix B implies. Two further checks separate coding error from algebra: the analytic derivatives match a complex-step reference to 7.8×10^{-9} , and a deterministic audit passes 82 of 82 conditional checks with exact-zero differences and 48 of 48 marginal checks with differences no larger than about 3.6×10^{-15} (Appendix E). These checks confirm that the Assumption 1 boundary holds in computation as well as in the algebra. The $\rho = 0.8$ runs confirm the same cross-interpretation identity at a representative alternative working correlation; they correspond to a different working model, reported as such and never pooled with the $\rho = 0$ headline.

Table 1 reports the target layer, and by design it gives two standard errors per coefficient: the profile standard error, which is the practical $\mathbf{H}_{\beta\beta}$ -bread estimator that standard software returns, and the full- ξ marginal standard error, which is the Definition 2 target with Schur bread, together with the signed percentage gap between them. In the Clinton synthesis the gap reaches +1.50% (the self-paced coefficient); in the Roberts synthesis it reaches +7.63% (the VT task coefficient). The gap measures

Theorem 1: META and SURVEY interpretations share one marginal likelihood ($\rho = 0$)

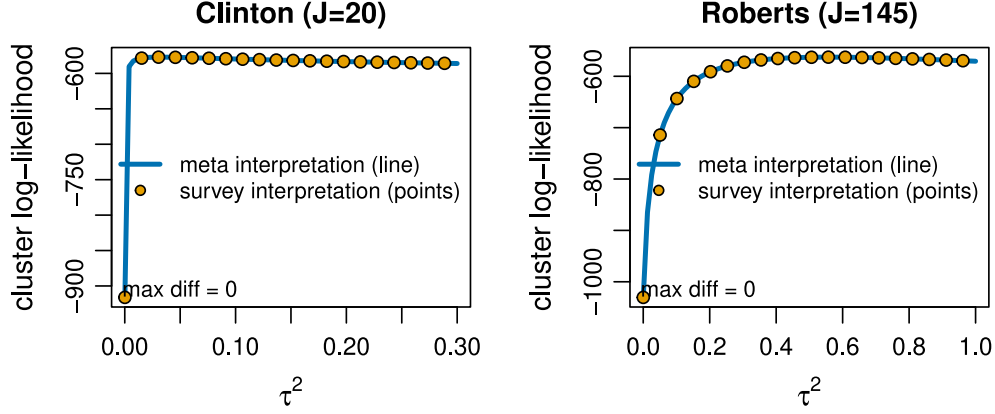


FIGURE 2. Likelihood identity on the two real designs. The marginal cluster Gaussian log likelihood $\ell^{\text{marg}}(\tau^2)$ is shown at $\rho = 0$ (the fitted correlated-effects working boundary) for the Clinton synthesis ($J = 20$, degenerate Case B) and the Roberts synthesis ($J = 145$, non-degenerate Case A). The meta-analytic and survey interpretations evaluate the same function to machine precision, $\max |\Delta| = 0$, at $\rho = 0$ (shown) and at $\rho = 0.8$ (Appendix F).

the τ^2 -propagation effect directly, and it behaves as the cross-information diagnostic predicts: it is induced by $\mathbf{H}_{\beta\tau^2}$ through the Schur bread, it is largest for the within-study-varying Roberts VT cell, and it would vanish under exact Cox–Reid orthogonality. At the score root, $\max |\mathbf{H}_{\beta\tau^2}| = 175.6$ (Clinton) and 15.5 (Roberts), with scale-free norms 0.21 and 0.06, so the diagnostic can be compared across two very differently sized designs. The recoloring layer then performs as Corollary 1 requires: the profile CCC standard error equals the `bayesRVE` sandwich standard error to 2.1×10^{-16} (Clinton) and 3.5×10^{-16} (Roberts), with the posterior mean untouched (Clinton intercept -0.581 ; Roberts EPT 0.402). Figure 3 displays the recoloring for a selected coefficient pair in each synthesis. The CR2-with-Satterthwaite small-sample layer is applied on top of the calibrated block and is reported in the table note, separate from the CR0 target it refines.

An external comparator provides an independent check on the target values. Re-fitting the same $\rho = 0$ study-mean working model independently and applying the CR2 correction of the `clubSandwich` package (Pustejovsky, 2026) reproduces the calibrated standard errors to 1.3% (Clinton) and 0.4% (Roberts); the residual reflects the expected difference between small-sample plug-in conventions and is largest in

TABLE 1. Selected fixed-effect coefficients for the Clinton reading-performance and Roberts enactment syntheses under the Bayesian correlated-effects model at $\rho = 0$ (the fitted working-likelihood boundary). The reported point estimate (*Bayes mean*) is the posterior mean $\bar{\xi}_{\text{post}}$, the affine-recoloring fixed point (center preserved, Corollary 1); the standard errors are the marginal Godambe sandwich evaluated at the sample root $\hat{\xi}$ (the Section 4 evaluation point). *Profile SE*: the practical β -block standard error with $H_{\beta\beta}$ bread (τ^2 held fixed). *Full- ξ marginal SE*: the standard error with Schur-complement bread $S = H_{\beta\beta} - H_{\beta\tau^2}H_{\tau^2\tau^2}^{-1}H_{\tau^2\beta}^\top$ (the theoretical target, which propagates τ^2 uncertainty). Both are likelihood-only (CR0). *Gap (%)* is the signed relative difference $100 \times (\text{Full-}\xi - \text{Profile})/\text{Profile}$, the τ^2 -propagation effect. The cluster-robust CR2 small-sample layer is in the Note.

Coefficient	Bayes mean	Profile SE	Full- ξ marginal SE	Gap (%)
<i>Clinton reading-performance synthesis ($J = 20$)</i>				
intercept	-0.581	0.1054	0.1067	+1.25
self-paced	0.396	0.1038	0.1054	+1.50
screen	0.074	0.0957	0.0956	-0.10
<i>Roberts enactment synthesis ($J = 145$)</i>				
EPT	0.402	0.1142	0.1147	+0.42
IT	0.514	0.1604	0.1616	+0.70
VT	1.276	0.0712	0.0767	+7.63

Note. The marginal target is the selected β -block of the full- ξ sandwich, $V_\xi^{\text{marg}}[\beta, \beta] = [H_\xi^{-1}M_\xi H_\xi^{-1}]_{\beta, \beta}$ with $\xi = (\beta, \tau^2)$, the meat M_ξ built from likelihood-only marginal cluster scores, and Schur-complement bread S , evaluated at the sample root $\hat{\xi}$ (the posterior mean $\bar{\xi}_{\text{post}}$ is the reported center; under regularity conditions both $\hat{\xi}$ and $\bar{\xi}_{\text{post}}$ are consistent for the population pseudo-true root $\xi^\dagger$). The β -block bread S propagates τ^2 estimation uncertainty through the cross-block $H_{\beta\tau^2}$; it is not a fixed- τ^2 object. **Interpreting the gap.** The gap is induced by the information cross-block $H_{\beta\tau^2}$, which propagates τ^2 estimation uncertainty into the β -block variance through the Schur bread S ; it is largest for the within-study-varying Roberts VT cell (+7.63%), and under exact orthogonality ($H_{\beta\tau^2} = 0$) it would vanish. **Small-sample layer.** Applying the cluster-robust CR2 correction with Satterthwaite degrees of freedom (at the posterior mean) gives the reportable CCC standard error (CCC-SD) = 0.115/0.114/0.105 (df 7.1/11.2/9.0; Clinton) and 0.118/0.168/0.072 (df 40.7/16.3/137.6; Roberts), the practical interval-width layer, separate from the CR0 target. **Orthogonality diagnostic.** $\max |H_{\beta\tau^2}| = 175.6$ (Clinton) and 15.5 (Roberts); the scale-free cross-information norm $\|H_{\beta\beta}^{-1/2}H_{\beta\tau^2}H_{\tau^2\tau^2}^{-1/2}\|_2 = 0.21$ (Clinton) and 0.06 (Roberts), comparable across the two designs.

Selected β -block CCC: covariance $\rightarrow$ fixed-effect sandwich target, center preserved

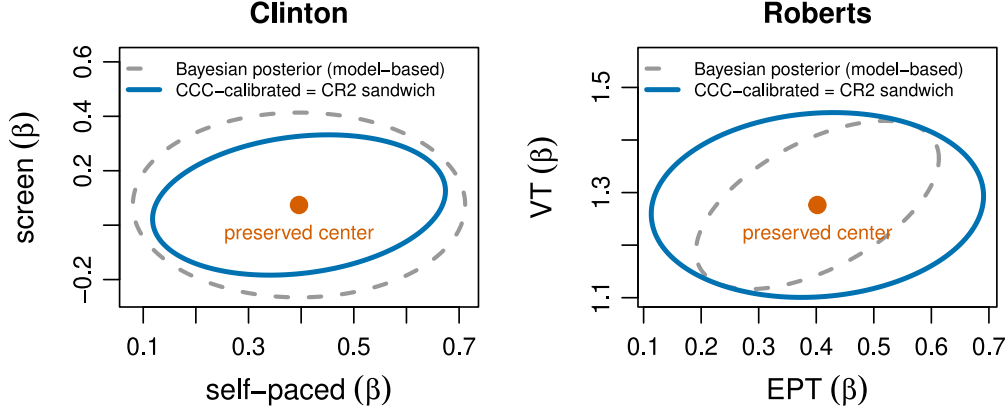


FIGURE 3. Covariance recoloring on the two real designs (Corollary 1). The model-based Bayesian posterior covariance (dashed) is recolored to the cluster-robust sandwich (solid), with the center preserved (filled point). The theorem-level target is the full- ξ CR0 marginal sandwich; the solid ellipse shows the optional CR2 reporting layer applied to it. Left: Clinton synthesis (Case B, study-constant moderators), where the sandwich is tighter than the model-based posterior. Right: Roberts synthesis (Case A, within-study-varying VT moderator), where the recoloring also reorients the covariance. Only the selected β -block is recolored, and the posterior mean does not move. Standard errors and the profile-versus-full- ξ gap appear in Table 1.

the $J = 20$ Clinton design. The published $\rho = 0.8$ observation-level benchmark of Pustejovsky and Chen (2024) is recorded in Appendix F as a labeled different-model comparator, not as a parity claim.

The two designs illustrate the estimand-existence point that motivated the pairing. In the Clinton synthesis (Case B), aggregation is lossless, the aggregation-route analysis of Pustejovsky and Chen (2024) coincides with the studies-as-clusters analysis, and the target confirms what either representation would report. In the Roberts synthesis (Case A), literal study-level averaging would erase the within-study task contrast, yet the bridge identity holds, the marginal target remains defined, and the calibrated covariance is delivered without aggregation or subgrouping machinery. On real data, then, the two interpretations produce the same numbers to machine precision, the calibrated standard error matches the marginal target, and the target remains available for the non-subgroupable moderators for which no aggregated representation exists. Reproducibility details and code pointers are collected in Appendix G.

6. Discussion

The most important result of this article is that the reporting problem for Bayesian dependent-effects meta-analysis has a definite answer. Once the correlated-effects working model is fit, the posterior center is determined, and the only open reporting object is the fixed-effect covariance block. The covariance that block should carry is the marginal fixed-effect sandwich of the studies-as-clusters estimand, with likelihood-only bread and meat and with the Schur-complement bread that carries the heterogeneity uncertainty, and an affine recoloring moves the posterior covariance onto that target while leaving the posterior mean fixed. Neither part of this answer was available from meta-analysis alone. The target is named by the survey calibration framework of M. R. Williams and Savitsky (2021), and the license to import it is the pair of identities in Theorems 1 and 2, under which the two interpretations fit one matched clustered Gaussian model with equal posteriors under matched priors and equal marginal sandwiches, together with the center-preserving map of Corollary 1 that attains the target exactly.

A second contribution is the transport map itself, that is, an explicit record of what carries across the bridge and under what conditions. The likelihood, the posterior measure under matched priors, the marginal fixed-effect sandwich target, and the recoloring transport as theorems. The conditional latent-effect sandwich and the aligned HC/CR small-sample corrections transport only under matched conventions, meaning the same data, designs, working correlation, and covariances on both sides (Proposition 3 and Corollary 2, Appendix D); they are not theorems at the breadth of Assumption 1. Full-joint conditional calibration does not transport at all, because Lemma 1 shows that its meat is rank-deficient. For the working meta-analyst, the practical implication is that the block of Definition 2 should be reported, with the orthogonality diagnostics used to check whether the profile standard error already coincides with the target and the recoloring applied when it does not.

The relation to Pustejovsky and Chen (2024) is complementary. Their equivalences relate meta-analytic representations to one another, running from aggregated to unaggregated forms through proportional likelihoods and matched-prior posterior coincidence, and they establish that reductionist strategies correspond to multivariate working models. The present bridge adds an interpretation from outside meta-analysis and applies it to the reporting problem. The survey interpretation contributes the covariance target that no within-meta-analysis equivalence could name, because the target belongs to the calibration literature rather than to the model family. Where

their aggregation route exists (Case B designs), the two frameworks produce the same numbers; where it does not (Case A designs with within-study moderator variation), the studies-as-clusters target keeps the estimand and its covariance defined.

The scope of the results is limited in several respects. First, the identity is proved on the matched unit-weighted clustered Gaussian boundary of Assumption 1, which requires unit weights at both the observation and the cluster level, a matched equicorrelated within-cluster covariance with known per-cluster scales, and a fixed conditional working correlation. The unit-weight requirement is less restrictive than it may appear, because a standard Bayesian fit already sits on this boundary: every cluster’s likelihood contribution enters the posterior with exponent one, and non-unit weights arise only when a pseudo-posterior construction reweights those contributions for an informative design. Effect-specific heteroscedastic within-study variances and informative, non-unit-weight designs both lie outside this boundary; the survey side already possesses pseudo-likelihood machinery for informative designs (León-Novelo & Savitsky, 2023), and extending the bridge to non-unit weights is the natural next step. Second, the results establish which covariance is reported and prove that the recoloring attains it; they make no finite-sample frequentist coverage claim, because coverage depends on the confidence-region procedure layered on the calibrated covariance. The operating characteristics of the full calibrated report, including interval coverage, positive-definiteness failure rates, and behavior across J and moderator structures, are accordingly the subject of the companion paper’s simulation study rather than of the present theory. Third, whether calibration materially changes the reported uncertainty in a given synthesis is an empirical question, and the two diagnostics of Section 4 are designed to answer it for the synthesis at hand. In the two illustrations the first-order correction, from the model-based posterior spread to the target, is substantial: the posterior standard deviations differ from the full- ξ target by roughly 15 to 45 percent with no consistent sign. The Clinton posterior is wider than the target, because the variance-component prior widens a $J = 20$ posterior, while the Roberts posterior is narrower, so reporting it directly would understate the robust uncertainty (Appendix F, Tables F.3 and F.4). The profile-versus-full- ξ gap of Table 1 is the smaller second-order refinement on top of that correction. The exactness of the results depends on these boundary conditions, and relaxing any of them requires new theory.

The present article and its companion divide the work as follows. This article establishes the bridge and the target: the likelihood identity, the posterior-measure

equality, the marginal sandwich of Definition 2, the rank obstruction, and the exact center-preserving recoloring. The companion paper develops the operational method built on that foundation: a three-layer calibration procedure with finite-sample CR2 matching and Satterthwaite degrees of freedom, its software implementation, and a simulation study of coverage across realistic synthesis designs. Analysts who want the method can use the `bayesRVE` package directly; the present article explains why the method targets the marginal sandwich and when that targeting is exact.

Dependent effect sizes are routine in educational and behavioral synthesis, and Bayesian working models for them are attractive because they propagate uncertainty coherently and support direct probability statements about moderator effects. A shortcoming of Bayesian dependent-effects synthesis, however, has been the absence of a named covariance target and of a way to reach it without disturbing the estimates. Recognizing the correlated-effects model and the unit-weighted clustered survey model as one likelihood supplies both, and it connects two literatures whose tools solve complementary halves of the same reporting problem. For this reason, we suggest that Bayesian analyses of dependent effect sizes report the calibrated covariance of Definition 2, or the profile standard error where the diagnostics show the two already coincide, in the same routine way that cluster-robust standard errors are now reported in frequentist syntheses.

Software and reproducibility. The Bayesian correlated-effects fits, the marginal fixed-effect sandwich, and the center-preserving calibration were computed with the `bayesRVE` R package (Lee, 2026) (<https://github.com/joonho112/bayesRVE>; documentation at <https://joonho112.github.io/bayesRVE/>), which draws posterior samples through Stan via the `cmdstanr` interface. The external comparator uses the CR2 correction of the `clubSandwich` package (Pustejovsky, 2026). The two deterministic algebra audits of Appendix E run in base R and Python and read no study data. A replication package with the ordered analysis scripts, the two audits, and a rendered walkthrough is available at <https://github.com/joonho112/likelihood-bridge-replication> (companion guide at <https://joonho112.github.io/likelihood-bridge-replication/>); every number reported in Section 5 and Appendix F regenerates from it.

Data availability. The empirical illustration uses two published syntheses, the Clinton reading-performance synthesis (Clinton-Lisell, 2021) and the Roberts enactment

synthesis (Roberts et al., 2022), as distributed in the replication materials of Pustejovsky and Chen (2024) on the Open Science Framework (<https://osf.io/9u52e/>). The data belong to the original authors and to those materials and are not redistributed with our code.

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Online Supplement

A Likelihood Bridge for Bayesian Meta-Analysis With Dependent Effect Sizes

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Appendix A. Notation, the translation map, and the scope of the supplement

This appendix fixes the reference material the other appendices use: the symbol crosswalk between the two interpretations, the domain ledger of the restricted translation map $\mathcal{T}_A$, three clarifications that later proofs lean on (the unit-weight condition, the global-shift nonidentifiability, and special condition S1), a small set of positioning notes that go beyond the main text, and the attribution and scope statements for the supplement as a whole.

A.1. Symbol crosswalk and vector routing. The notation follows Sections 2–3. One symbol set carries both interpretations of the same clustered object; the crosswalk supplies vocabulary and asserts no equivalence beyond the matched boundary.

TABLE A.1. Symbol crosswalk between the meta-analytic and survey interpretations.

Symbol	Common object	Meta-analysis interpretation	Survey interpretation
j	independent cluster	study	primary sampling unit / sampled cluster
i	unit within cluster	effect-size estimate	sampled observation
n_j	cluster size	effect sizes in study j	sampled units in cluster j
$\mathbf{y}_j$	response vector	effect-size estimates	survey responses
$\mathbf{X}_j$	$n_j \times p$	effect-size-level	observation-level
	observation-level design	moderators	covariates
β	fixed effects	meta-regression coefficients	regression coefficients
θ_j	latent cluster effect	study-specific deviation	cluster random effect
τ^2	between-cluster variance	between-study heterogeneity	between-cluster variance
a_j	known within-cluster scale	common within-study scale $\bar{s}_j^2$	common within-cluster scale $\sigma_{e,j}^2$
ρ	fixed working correlation	within-study working correlation	within-cluster conditional residual correlation
$\tilde{w}_{ij}, \tilde{w}_j$	normalized weights	implicit unit weights	survey weights, fixed to 1

The shared conditional model, its equicorrelated specialization, and the three parameter vectors are

$$\mathbf{y}_j \mid \beta, \theta_j \sim \mathbf{N}_{n_j}(\mathbf{X}_j \beta + \theta_j \mathbf{1}_{n_j}, \mathbf{V}_j), \quad \theta_j \mid \tau^2 \sim \mathbf{N}(0, \tau^2),$$

$$\begin{aligned}\mathbf{V}_j &= a_j \mathbf{C}_j(\rho), & \mathbf{C}_j(\rho) &= (1 - \rho) \mathbf{I}_{n_j} + \rho \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top, & a_j &> 0, \\ \boldsymbol{\eta} &= (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top)^\top, & \boldsymbol{\zeta} &= (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top, \tau^2)^\top, & \boldsymbol{\xi} &= (\boldsymbol{\beta}^\top, \tau^2)^\top,\end{aligned}$$

with independent clusters, $\boldsymbol{\theta} = (\theta_1, \dots, \theta_J)^\top$, and, for $n_j > 1$, positive definiteness of $\mathbf{C}_j(\rho)$ on $-1/(n_j - 1) < \rho < 1$. The rows of $\mathbf{X}_j$ may vary within cluster; the design is never silently replaced by cluster means. The routing of the three vectors matters throughout and is preserved in every appendix: $\boldsymbol{\eta}$ indexes the conditional latent-effect likelihood and its score, Hessian, and sandwich objects; $\boldsymbol{\zeta}$ indexes posterior-measure and MAP statements, because τ^2 belongs to the hierarchical posterior; and $\boldsymbol{\xi}$ indexes the marginal fixed-effect likelihood after the latent effects are integrated out, which is the space of the reported target. A sandwich built in $\boldsymbol{\eta}$ excludes τ^2 and is therefore a conditional latent-effect object, not a full-joint sandwich. The reported quantity is a selected fixed-effect block or contrast $\boldsymbol{\psi} = \mathbf{R}_\psi \boldsymbol{\beta}$, and its covariance target is the selected $\boldsymbol{\beta}$ -block of the marginal $\boldsymbol{\xi}$ -sandwich (Definition 2; Appendix C).

A.2. Domain ledger of the restricted translation map. The correspondence between the two interpretations is a restricted translation map $\mathcal{T}_A$, defined on a sharply delimited domain and false outside it. Each row of the ledger states what the corresponding ingredient is defined for and undefined for; the map holds as an identity only when every row is satisfied simultaneously.

TABLE A.2. Domain ledger for $\mathcal{T}_A$.

Ingredient	Defined for	Undefined for
Cluster partition	the same independent cluster partition	different partitions; merged or split clusters
Ordered responses	the same ordered $\mathbf{y}_j$	changed observations or within-cluster ordering
Design matrices	the same observation-level $\mathbf{X}_j$	changed designs; cluster-mean aggregation
Working correlation	fixed matched admissible ρ	estimated or posterior-integrated ρ
Known covariance	matched known positive-definite $\mathbf{V}_j$	unmatched covariances
Common scale	$\bar{s}_j^2 = \sigma_{e,j}^2 = a_j$ with $\mathbf{V}_j = a_j \mathbf{C}_j(\rho)$	scale mismatch; effect-specific heteroscedastic variances

Ingredient	Defined for	Undefined for
Weights	unit-normalized $\tilde{w}_{ij} = 1$ and $\tilde{w}_j = 1$	arbitrary weights; non-unit equal weights; tempered likelihoods
Level-1 kernel	a shared Gaussian kernel	non-Gaussian kernels
Random-effect law (marginal statements)	the same $\theta_j \mid \tau^2 \sim \mathbf{N}(0, \tau^2)$	different or interpretation-specific random-effect laws
Likelihood objects	observed-vector likelihood objects under the stated Gaussian parameterization	finite-population survey estimands not represented by the same likelihood object
Priors (posterior statements)	matched priors, common support, propriety	different priors, supports, or parameterizations

Two clarifications qualify how the ledger should be read. The matched-covariance row is recorded at two levels: the underlying algebra would already apply to any matched known symmetric positive-definite matrix, but the bridge established in this paper is for the homoscedastic equicorrelated specialization $\mathbf{V}_j = a_j \mathbf{C}_j(\rho)$; relaxing a_j to effect-specific within-cluster variances is future work. And the design-matrix row is unaggregated: both interpretations retain the same observation-level $\mathbf{X}_j$, so within-cluster covariate variation remains in scope and the map licenses no cluster-mean reduction.

A.3. Unit-normalized weights versus equal non-unit weights. The weight condition is $\tilde{w}_{ij} = 1$ and $\tilde{w}_j = 1$, which is stronger than “all weights are equal,” and the distinction changes the inferential object. Multiplying every cluster contribution by a non-unit constant may leave a likelihood-only maximizer unchanged, but the posterior is tempered relative to the prior: a posterior proportional to $\exp\{\ell(\boldsymbol{\zeta})\} \pi(\boldsymbol{\zeta})$ is in general a different probability measure from

$$\exp\{c \ell(\boldsymbol{\zeta})\} \pi(\boldsymbol{\zeta}), \quad c \neq 1,$$

so posterior centers, MAP sets under prior curvature, posterior covariances, and the pseudo-score geometry that defines the transported target all move. Equal non-unit weights therefore constitute a different inferential regime rather than a harmless

rescaling, and arbitrary informative weights a further regime again; both are future-work directions on the survey side rather than corollaries of the present result.

A.4. Global-shift nonidentifiability. When $\mathbf{X}_j$ contains a global intercept β_0 while each cluster carries a free latent effect, the conditional likelihood is flat along a global-shift direction: for any constant c ,

$$\beta'_0 = \beta_0 + c, \quad \theta'_j = \theta_j - c,$$

leaves every cluster mean unchanged,

$$\beta'_0 \mathbf{1}_{n_j} + \theta'_j \mathbf{1}_{n_j} = \beta_0 \mathbf{1}_{n_j} + \theta_j \mathbf{1}_{n_j},$$

and hence leaves the residuals and the conditional log likelihood unchanged. Both interpretations carry exactly the same flat direction, so the nonidentifiability cannot separate the two likelihoods and does not affect Theorem 1. It does matter for MAP uniqueness, inverse-Hessian arguments, and conditional sandwich objects, which need prior curvature, an identifying constraint, or an estimable-subspace convention; those conventions are recorded in Appendix D. It is also one reason the reported target is routed through the marginal $\boldsymbol{\xi}$ model: integrating out the latent effects removes the flat direction and lowers the dimension from $p + J$ to $p + 1$.

A.5. Special condition S1. A cluster-constant design, $\mathbf{x}_{ij} = \mathbf{x}_j$ for all i so that $\mathbf{X}_j = \mathbf{1}_{n_j} \mathbf{x}_j^\top$, is labeled special condition S1. S1 is not part of Assumption 1 and is required by none of the identities; it matters only for closed-form cluster-mean or aggregation-style simplifications that this paper does not use. Proposition 1 of Appendix B establishes this.

A.6. Positioning notes beyond the main text. Three positioning clarifications exceed what Section 2 could carry and are recorded here.

The survey interpretation is model-assisted, not a literal design. The paper does not claim that meta-analytic studies are sampled under a known design with inclusion probabilities. It posits a super-population of study objects, treats included studies as independent primary sampling units, and observes all effect-size cells within each included study. Under the usual existence and estimability conditions the marginal fixed effect $\boldsymbol{\beta}$ is then a genuine estimand in Binder’s implicit-parameter sense (the solution of a population estimating equation in the working metric), even though the within-cluster covariance is only a working covariance (Binder, 1983). On the

unit-weight boundary the weights are inert by construction; their role is to identify the shared object, not to alter it. The survey-side program that motivates this stance frames an *approximate* bridge between pseudo-Bayesian mixed models and clustered inference (Savitsky & Williams, 2022); the present paper proves one exact unit-weighted boundary case adjacent to that program.

The result is not a composite-likelihood repair. Composite-likelihood procedures start from a surrogate objective for an intractable joint likelihood; the surrogate is misspecified by construction, the second Bartlett identity fails, and the Godambe sandwich recovers what the surrogate discarded (Chen et al., 2015; Varin, 2008; Varin et al., 2011). Here there is no surrogate: the full matched Gaussian likelihood is the object under analysis, and the two interpretations evaluate the same function. What the construction does *not* assume is correctness of the working model relative to the data-generating process: $\mathbf{V}_j$ and the Gaussian kernel are working approximations, Bartlett’s second identity need not hold under that misspecification, and the marginal Godambe sandwich supplies the misspecification-robust fixed-effect covariance accordingly. What separates the present construction from the composite-likelihood apparatus is the absence of a surrogate objective, not an assumption of correct specification. The nearest mechanical neighbor samples from a working-independence composite posterior for network meta-analysis and applies an open-faced-sandwich map that rescales the whole parameter vector (Wang et al., 2025); that operation repairs an over-precise composite posterior, whereas the present calibration realigns only the selected β -block of a matched-model posterior that is not being repaired. Adjacent network, bounded-influence, and diagnostic-accuracy variants use approximate or robustified objectives for different model classes and are contrasts only (Guolo, 2023; Hu et al., 2025; Liu et al., 2025); and the transported object is the model-based marginal fixed-effect sandwich carrying τ^2 as a nuisance coordinate, not a generic working-independence sandwich of the generalized-estimating-equations tradition (Liang & Zeger, 1986).

Relationship to Pustejovsky and Chen (2024), long form. Pustejovsky and Chen (2024) prove that an inverse-variance-aggregated meta-regression and the correlated-effects model have *proportional* likelihoods, record the one-line corollary that matched priors give coinciding Bayesian posteriors, and extend the equivalence to robust variance estimation, including a CR2 matrix equality that requires additional compound-symmetry and balance structure (Tipton, 2015). The present paper changes both the

object and the payoff: the object moves from proportionality between two meta-analytic representations to a measure identity of one matched Gaussian model under two paradigms, with zero additive constant, and the payoff moves from lossless aggregation to the design-based fixed-effect covariance target with its center-preserving calibration. On the matched common-scale boundary their CR2 equality is reproduced exactly rather than contested; this agreement shows that the design interpretation generalizes their result rather than colliding with it. Off that boundary the decisive difference is estimand existence: one-row-per-study aggregation, confined to study-level predictors, can lack a left-hand side when a within-study moderator is not amenable to categorical subgrouping (their subgrouping and shifting-unit constructions do cover categorical within-study moderators), whereas the unaggregated studies-as-clusters estimand and its marginal sandwich remain defined without any subgrouping requirement.

A.7. Provenance and attribution. The attribution boundary of the contribution is recorded here. Imported: the center-preserving calibration principle and its survey-side blueprint (Little, 2012; Rubin, 1984; M. R. Williams & Savitsky, 2021), the design-based estimating-equation frame (Binder, 1983), the RVE and cluster-sandwich vocabulary in which the target is written, and the proportional-likelihood equivalence genre within meta-analysis (Pustejovsky & Chen, 2024). New in this paper: the exact unit-weighted likelihood identity and its posterior-measure consequence (Theorem 1, Proposition 2), the identification of the marginal fixed-effect Godambe sandwich as the covariance target of the Bayesian correlated-effects report (Theorem 2 with Definition 2), the rank obstruction that closes the full-joint conditional route (Lemma 1), and the finite-sample exactness of the center-preserving recoloring (Corollary 1). Software, data, and comparator citations throughout the supplement are provenance only: no software output, package default, or comparator agreement is used as proof, definition, or validation of any result.

A.8. Scope of the supplement. One scope statement governs every appendix and is not repeated in them. The results are conceptual rather than procedural: no meta-analysis is routed through survey software, and no claim is made that software defaults agree. The calibration preserves the posterior center and realigns only the reported covariance of the selected block. Nothing here is a frequentist coverage theorem; aligning the reported covariance fixes which object is reported, and coverage depends on the interval procedure layered on top. Posterior agreement is equality of probability

measures, never a claim about finite Markov chain Monte Carlo draws. The identity does not cover non-unit or informative weights, unmatched or effect-specific heteroscedastic covariances, estimated or posterior-integrated ρ , non-Gaussian kernels, or correlated-and-hierarchical-effects extensions; these are the future-work boundary stated in Section 6. The appendices that follow prove the identities (B), derive the target and the calibration (C), record the matched-convention supporting results (D), audit the algebra on a finite design (E), realize the results on two published syntheses (F), and document reproducibility (G).

Appendix B. Proof of Theorem 1 and its consequences

This appendix proves the clusterwise likelihood identity of Theorem 1, derives the two boundary caveats, and proves Propositions 1 and 2. Displays carry appendix-local numbers (B.k); cross-document pointers are written in prose.

B.1. Setup. Clusters are indexed by $j = 1, \dots, J$, units within cluster by $i = 1, \dots, n_j$, and $N = \sum_{j=1}^J n_j$. The vector $\mathbf{1}_{n_j}$ is the n_j -vector of ones and $\mathbf{I}_{n_j}$ the $n_j \times n_j$ identity. The response vector $\mathbf{y}_j \in \mathbb{R}^{n_j}$ and the observation-level design matrix $\mathbf{X}_j \in \mathbb{R}^{n_j \times p}$ are common to the two interpretations, and the rows of $\mathbf{X}_j$ may vary within cluster. Superscripts M and S denote the meta-analytic and survey interpretations and are written only when a distinction is needed.

Conditional on the fixed effects $\boldsymbol{\beta} \in \mathbb{R}^p$ and the latent cluster effect θ_j , the shared working model is

$$\mathbf{y}_j \mid \boldsymbol{\beta}, \theta_j \sim \mathbf{N}_{n_j}(\mathbf{X}_j \boldsymbol{\beta} + \theta_j \mathbf{1}_{n_j}, \mathbf{V}_j), \quad \theta_j \mid \tau^2 \sim \mathbf{N}(0, \tau^2), \quad (\text{B.1})$$

with independent clusters. The cluster residual and the clusterwise conditional Gaussian log likelihood are

$$\mathbf{r}_j(\boldsymbol{\eta}) = \mathbf{y}_j - \mathbf{X}_j \boldsymbol{\beta} - \theta_j \mathbf{1}_{n_j}, \quad (\text{B.2})$$

$$\ell_j(\boldsymbol{\eta}) = -\frac{1}{2} \{n_j \log(2\pi) + \log |\mathbf{V}_j| + \mathbf{r}_j(\boldsymbol{\eta})^\top \mathbf{V}_j^{-1} \mathbf{r}_j(\boldsymbol{\eta})\}, \quad (\text{B.3})$$

with total conditional log likelihood $\ell(\boldsymbol{\eta}) = \sum_{j=1}^J \ell_j(\boldsymbol{\eta})$. The three parameter vectors are

$$\boldsymbol{\eta} = (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top)^\top, \quad \boldsymbol{\zeta} = (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top, \tau^2)^\top, \quad \boldsymbol{\xi} = (\boldsymbol{\beta}^\top, \tau^2)^\top, \quad (\text{B.4})$$

with $\boldsymbol{\theta} = (\theta_1, \dots, \theta_J)^\top$; their routing is fixed in Appendix A. Integrating θ_j out of (B.1) gives the marginal covariance and model

$$\boldsymbol{\Sigma}_j(\tau^2) = \mathbf{V}_j + \tau^2 \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top, \quad \mathbf{y}_j \mid \boldsymbol{\beta}, \tau^2 \sim \mathbf{N}_{n_j}(\mathbf{X}_j \boldsymbol{\beta}, \boldsymbol{\Sigma}_j(\tau^2)), \quad (\text{B.5})$$

used in Sections B.4 and B.5 and taken up in Appendix C. The covariance specialization is the equicorrelated common-scale form

$$\mathbf{V}_j = a_j \mathbf{C}_j(\rho), \quad \mathbf{C}_j(\rho) = (1 - \rho) \mathbf{I}_{n_j} + \rho \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top, \quad a_j > 0. \quad (\text{B.6})$$

For $n_j > 1$, positive definiteness of $\mathbf{C}_j(\rho)$ holds on $-1/(n_j - 1) < \rho < 1$; the working correlation ρ is fixed, admissible, and matched across the two interpretations. A singleton cluster ($n_j = 1$) is the harmless special case $\mathbf{C}_j(\rho) = 1$, $\mathbf{V}_j = a_j$, in which (B.1) reduces to the univariate random-effects model and no separate treatment is needed. The conditions used throughout are Assumption 1 (A1–A7) of Section 3, identified through the restricted translation map $\mathcal{T}_A$, whose domain ledger is Appendix A; the weight condition $\tilde{w}_{ij} = 1$, $\tilde{w}_j = 1$ is unit normalization, not “equal weights up to scale” (Appendix A.3). Special condition S1 (cluster-constant design) is not part of Assumption 1; its non-core status is Proposition 1, proved in Section B.5.

B.2. Matched compound-symmetric covariance geometry. The likelihood identity is cleanest at the matched positive-definite level: if both interpretations use the same known symmetric positive-definite $\mathbf{V}_j$, the determinant, inverse, residual, and quadratic form in (B.3) are the same function of $\boldsymbol{\eta}$ in both interpretations, and nothing in that statement uses compound symmetry. The substantive boundary of the paper is narrower, fixing $\mathbf{V}_j = a_j \mathbf{C}_j(\rho)$ as in (B.6); this section records the determinant, inverse, and quadratic-form algebra of that specialization, and Section B.3 uses it to locate where a discrepancy could in principle arise and to show that it does not.

Eigenvalue split. The rank-one matrix $\mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top$ has eigenvalue n_j along $\mathbf{1}_{n_j}$ and eigenvalue 0 on the $(n_j - 1)$ -dimensional subspace orthogonal to $\mathbf{1}_{n_j}$. Substituting into $\mathbf{C}_j(\rho) = (1 - \rho) \mathbf{I}_{n_j} + \rho \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top$ gives the eigenvalues of $\mathbf{C}_j(\rho)$,

$$\begin{aligned} \lambda_{\text{within}} &= 1 - \rho, & \text{with multiplicity } n_j - 1, \\ \lambda_{\text{between}} &= 1 + (n_j - 1)\rho, & \text{with multiplicity } 1, \end{aligned} \quad (\text{B.7})$$

the between-cluster eigenvalue attached to the direction $\mathbf{1}_{n_j}$ and the within-cluster eigenvalue attached to the contrast subspace. The labels name the eigenvector directions, not the relative magnitudes: for $\rho < 0$ the between-cluster eigenvalue is the smaller of the two. Multiplying by a_j scales the eigenvalues to $a_j(1 - \rho)$ and $a_j[1 + (n_j - 1)\rho]$ and leaves the eigenvectors unchanged; positive definiteness of $\mathbf{V}_j$ is exactly $1 - \rho > 0$ and $1 + (n_j - 1)\rho > 0$.

Log-determinant. Because $|\mathbf{V}_j|$ is the product of its eigenvalues raised to their multiplicities,

$$\begin{aligned} \log |\mathbf{V}_j| &= (n_j - 1) \log[a_j(1 - \rho)] + \log[a_j\{1 + (n_j - 1)\rho\}] \\ &= n_j \log a_j + (n_j - 1) \log(1 - \rho) + \log[1 + (n_j - 1)\rho]. \end{aligned} \quad (\text{B.8})$$

Inverse. Applying the Sherman–Morrison formula to $\mathbf{V}_j = a_j[(1 - \rho)\mathbf{I}_{n_j} + \rho \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top]$, with base $a_j(1 - \rho)\mathbf{I}_{n_j}$ and rank-one update $a_j \rho \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top$,

$$\mathbf{V}_j^{-1} = \frac{1}{a_j(1 - \rho)} \left[\mathbf{I}_{n_j} - \frac{\rho}{1 + (n_j - 1)\rho} \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top \right]. \quad (\text{B.9})$$

Equivalently, (B.9) is the spectral inverse with eigenvalue $1/[a_j(1 - \rho)]$ on the contrast subspace and $1/\{a_j[1 + (n_j - 1)\rho]\}$ along $\mathbf{1}_{n_j}$. The denominator is $1 + (n_j - 1)\rho$, never $1 + n_j\rho$, and the scale a_j multiplies the whole inverse; both are carried exactly below.

Residual quadratic form. Using (B.9) and $\mathbf{1}_{n_j}^\top \mathbf{r}_j \mathbf{1}_{n_j}^\top \mathbf{r}_j = (\mathbf{1}_{n_j}^\top \mathbf{r}_j)^2$,

$$\mathbf{r}_j(\boldsymbol{\eta})^\top \mathbf{V}_j^{-1} \mathbf{r}_j(\boldsymbol{\eta}) = \frac{1}{a_j(1 - \rho)} \left[\mathbf{r}_j(\boldsymbol{\eta})^\top \mathbf{r}_j(\boldsymbol{\eta}) - \frac{\rho}{1 + (n_j - 1)\rho} \{\mathbf{1}_{n_j}^\top \mathbf{r}_j(\boldsymbol{\eta})\}^2 \right]. \quad (\text{B.10})$$

Reading (B.10) through the eigen-split, with $\bar{r}_j = n_j^{-1} \mathbf{1}_{n_j}^\top \mathbf{r}_j(\boldsymbol{\eta})$ the residual cluster mean and r_{ij} the components of $\mathbf{r}_j(\boldsymbol{\eta})$, the orthogonal decomposition into the projection onto $\mathbf{1}_{n_j}$ and the within-cluster contrasts gives

$$\mathbf{r}_j(\boldsymbol{\eta})^\top \mathbf{V}_j^{-1} \mathbf{r}_j(\boldsymbol{\eta}) = \frac{1}{a_j(1 - \rho)} \sum_{i=1}^{n_j} (r_{ij} - \bar{r}_j)^2 + \frac{n_j \bar{r}_j^2}{a_j\{1 + (n_j - 1)\rho\}}, \quad (\text{B.11})$$

algebraically identical to (B.10). In every one of (B.8)–(B.11) the parameter vector $\boldsymbol{\eta}$ enters only through the residual $\mathbf{r}_j(\boldsymbol{\eta})$, while the within-cluster scale enters only through a_j and the correlation only through ρ .

B.3. Full proof of Theorem 1. Let $\ell_j^{\text{M}}(\boldsymbol{\eta})$ denote the meta-analytic interpretation of (B.3), evaluated with $\mathbf{V}_j = \bar{s}_j^2 \mathbf{C}_j(\rho)$, and let $\ell_j^{\text{S}}(\boldsymbol{\eta})$ denote the survey interpretation:

the unit-weighted clusterwise Gaussian pseudo-log-likelihood with $\tilde{w}_{ij} = 1$, the unit-weight special case of the clustered survey pseudo-posterior recalled in Section 3, which therefore coincides with the full multivariate-normal log density evaluated at $\mathbf{V}_j = \sigma_{e,j}^2 \mathbf{C}_j(\rho)$. The total log likelihoods are $\ell^{\mathbf{R}}(\boldsymbol{\eta}) = \sum_{j=1}^J \ell_j^{\mathbf{R}}(\boldsymbol{\eta})$ for $\mathbf{R} \in \{\mathbf{M}, \mathbf{S}\}$.

Theorem 1 (restated; clusterwise likelihood identity). Let Assumption 1 hold and let the two interpretations be identified through the restricted translation map $\mathcal{T}_A$. Under the matched multivariate-normal normalization,

$$\ell_j^{\mathbf{M}}(\boldsymbol{\eta}) = \ell_j^{\mathbf{S}}(\boldsymbol{\eta}) \quad \text{for every cluster } j \text{ and every } \boldsymbol{\eta}, \quad (\text{B.12})$$

and therefore, summing over the independent clusters,

$$\ell^{\mathbf{M}}(\boldsymbol{\eta}) = \ell^{\mathbf{S}}(\boldsymbol{\eta}). \quad (\text{B.13})$$

The equality is exact: under the matched normalization the additive constant relating the two log likelihoods is zero. Under an alternative admissible normalization the two log likelihoods may differ, but only by a cluster-additive term that is free of every parameter being optimized, differentiated, or integrated.

Proof. The argument is cleanest at the matched positive-definite level and then specializes to the equicorrelated covariance.

Matched positive-definite level. By A2 the two interpretations use the same ordered response vector $\mathbf{y}_j$; by A3 the same observation-level design matrix $\mathbf{X}_j$; and both place the common fixed effects $\boldsymbol{\beta}$ and the same latent cluster effect θ_j in the linear predictor $\mathbf{X}_j\boldsymbol{\beta} + \theta_j\mathbf{1}_{n_j}$. Hence the cluster residual

$$\mathbf{r}_j(\boldsymbol{\eta}) = \mathbf{y}_j - \mathbf{X}_j\boldsymbol{\beta} - \theta_j\mathbf{1}_{n_j}$$

is the same vector-valued function of $\boldsymbol{\eta}$ in both interpretations. By A4–A6 the two interpretations use the same known positive-definite covariance matrix $\mathbf{V}_j$; in the present specialization $\mathbf{V}_j = a_j \mathbf{C}_j(\rho)$ with the per-cluster common scale $\bar{s}_j^2 = \sigma_{e,j}^2 = a_j$ and the common fixed ρ . Consequently the three ingredients of (B.3), namely the log-determinant $\log |\mathbf{V}_j|$, the inverse $\mathbf{V}_j^{-1}$, and the quadratic form $\mathbf{r}_j(\boldsymbol{\eta})^\top \mathbf{V}_j^{-1} \mathbf{r}_j(\boldsymbol{\eta})$, are identical in the two interpretations, as is the normalizing term $n_j \log(2\pi)$. By A7 the survey weights enter as $\tilde{w}_{ij} = 1$, so the clusterwise pseudo-log-likelihood reduces to the unweighted clusterwise Gaussian log likelihood of the meta-analytic interpretation and is not raised to a non-unit likelihood power. Therefore every term of (B.3) agrees term by term, giving $\ell_j^{\mathbf{M}}(\boldsymbol{\eta}) = \ell_j^{\mathbf{S}}(\boldsymbol{\eta})$ for every $\boldsymbol{\eta}$, which is (B.12). Because the clusters are independent in both interpretations,

$$\ell^{\mathbf{M}}(\boldsymbol{\eta}) = \sum_{j=1}^J \ell_j^{\mathbf{M}}(\boldsymbol{\eta}) = \sum_{j=1}^J \ell_j^{\mathbf{S}}(\boldsymbol{\eta}) = \ell^{\mathbf{S}}(\boldsymbol{\eta}),$$

which is (B.13).

Equicorrelated specialization. To locate where a discrepancy could in principle arise, and to see why it does not, specialize $\mathbf{V}_j = a_j \mathbf{C}_j(\rho)$ to the equicorrelated form and substitute the determinant (B.8), the inverse (B.9), and the quadratic form (B.10). In each of these the parameter vector $\boldsymbol{\eta}$ enters only through $\mathbf{r}_j(\boldsymbol{\eta})$, while the within-cluster scale enters only through a_j and the correlation only through ρ . The only quantities that could separate the interpretations are the scale-bearing factors in the determinant (B.8) and the leading coefficient in the quadratic form (B.10); both are matched by A6 (common a_j) and A4 (common ρ), and both are free of $\boldsymbol{\eta}$ in any case. The per-cluster additive constant is therefore exactly zero, not merely parameter-free.

Clusterwise multivariate form and ρ -dependent observationwise factorization. The survey pseudo-kernel display in Section 3 is the clusterwise multivariate Gaussian log likelihood. When $\rho = 0$, the matrix $\mathbf{V}_j = a_j \mathbf{I}_{n_j}$ is diagonal, so the quadratic form $\mathbf{r}_j^{\mathbf{T}} \mathbf{V}_j^{-1} \mathbf{r}_j = (1/a_j) \sum_{i=1}^{n_j} r_{ij}^2$ is a sum of univariate terms, and the multivariate log likelihood reduces to a sum of univariate log densities equivalent to a product of univariate densities $\prod_i p(y_{ij} | \dots)$. When $\rho > 0$, the off-diagonal entries of $\mathbf{C}_j(\rho)$ in (B.6) introduce cross-terms $r_{ij} r_{i'j}$ ($i \neq i'$) in the quadratic form (B.10), so the multivariate log likelihood does not factor into a sum of univariate contributions; the observationwise product notation is valid only at $\rho = 0$ and must be read as the full clusterwise multivariate form for $\rho > 0$. The identity of Theorem 1 holds for all $\rho \in (-1/(n_j - 1), 1)$.

Matched normalization. The “matched normalization” qualifier is used in its exact sense: both interpretations use the full Gaussian log density in (B.3), including the $n_j \log(2\pi)$ term and the full $\log |\mathbf{V}_j|$, so there is no leftover constant to absorb. If one interpretation were instead written under a different but admissible normalizing convention (for instance a kernel that omits $n_j \log(2\pi)$, or a weighted convention retaining a non-unit normalizing factor), the two log likelihoods would differ by a cluster-additive term free of $\boldsymbol{\eta}$, hence inert for estimation, differentiation, and integration. We state the result as equality because the normalization actually used is the matched one. $\square$

Under $\mathcal{T}_A$, the meta-analytic and survey labels do not generate two likelihoods; they name the same ordered data, residual vector, covariance matrix, Gaussian kernel, and unit-normalized weighting convention. In contrast to the aggregation- and reclassification-based equivalences of Pustejovsky and Chen (2024), neither interpretation changes the unit of analysis or aggregates the data, and the equality carries a zero constant rather than a proportionality factor.

B.4. Boundary caveat derivations. Theorem 1 is exact because its boundary conditions are exact. Two features of that exactness are easy to misread; each marks a boundary of $\mathcal{T}_A$ rather than a removable constant.

B.4.1. Caveat 1: scale mismatch is parameter-dependent. Caveat 1 (restated). The per-cluster scale match in A6, $\bar{s}_j^2 = \sigma_{e,j}^2 = a_j$, is essential. If the two interpretations use different cluster scales, so that $\mathbf{V}_j^M = \bar{s}_j^2 \mathbf{C}_j(\rho) \neq \sigma_{e,j}^2 \mathbf{C}_j(\rho) = \mathbf{V}_j^S$, then the clusterwise log likelihoods differ by a parameter-dependent quadratic rather than a constant, so the scores and curvature of the two interpretations no longer agree.

Derivation. Conditions A2–A4 still make the residual $\mathbf{r}_j(\boldsymbol{\eta})$ common to the two interpretations, and the $n_j \log(2\pi)$ terms still cancel. Differencing (B.3) leaves only the determinant and quadratic-form terms, so

$$\ell_j^M(\boldsymbol{\eta}) - \ell_j^S(\boldsymbol{\eta}) = -\frac{1}{2} \left[\log \frac{|\mathbf{V}_j^M|}{|\mathbf{V}_j^S|} + \mathbf{r}_j(\boldsymbol{\eta})^\top \{(\mathbf{V}_j^M)^{-1} - (\mathbf{V}_j^S)^{-1}\} \mathbf{r}_j(\boldsymbol{\eta}) \right]. \quad (\text{B.14})$$

For the equicorrelated common form the two interpretations share $\mathbf{C}_j(\rho)$ and differ only in the scalar scale. From (B.8), $\log |\mathbf{V}_j^M| - \log |\mathbf{V}_j^S| = n_j \log(\bar{s}_j^2 / \sigma_{e,j}^2)$, since the $\mathbf{C}_j(\rho)$ contribution cancels. From (B.9), $(\mathbf{V}_j^M)^{-1} - (\mathbf{V}_j^S)^{-1} = (\bar{s}_j^{-2} - \sigma_{e,j}^{-2}) \mathbf{C}_j(\rho)^{-1}$, where $\mathbf{C}_j(\rho)^{-1} = a_j \mathbf{V}_j^{-1}$ is the unit-scale correlation inverse. Hence

$$\ell_j^M(\boldsymbol{\eta}) - \ell_j^S(\boldsymbol{\eta}) = -\frac{1}{2} \left[n_j \log \frac{\bar{s}_j^2}{\sigma_{e,j}^2} + \left(\frac{1}{\bar{s}_j^2} - \frac{1}{\sigma_{e,j}^2} \right) \mathbf{r}_j(\boldsymbol{\eta})^\top \mathbf{C}_j(\rho)^{-1} \mathbf{r}_j(\boldsymbol{\eta}) \right]. \quad (\text{B.15})$$

The determinant portion is free of $(\boldsymbol{\beta}, \boldsymbol{\theta})$. The quadratic portion is not: whenever $\bar{s}_j^2 \neq \sigma_{e,j}^2$ and the residual is nonzero, it varies with $\boldsymbol{\eta}$ through $\mathbf{r}_j(\boldsymbol{\eta})$, and its $\boldsymbol{\eta}$ -gradient and $\boldsymbol{\eta}$ -Hessian are not identically zero. Scale mismatch therefore changes the parameter-dependent likelihood geometry: a single global survey error variance matched to studies with heterogeneous $\bar{s}_j^2$ breaks the identity at the score and Hessian level, not merely at the level of normalization. This is why A6 is a per-cluster condition and why scale mismatch lies outside the domain of $\mathcal{T}_A$. $\square$

B.4.2. *Caveat 2: the marginal determinant is not a removable constant.* **Caveat 2 (restated).** Theorem 1 concerns the conditional likelihood indexed by $\boldsymbol{\eta}$, in which the latent effects are held fixed and $\mathbf{V}_j$ is known. When the cluster effects are integrated out, the relevant object is the marginal model (B.5) with covariance $\boldsymbol{\Sigma}_j(\tau^2) = \mathbf{V}_j + \tau^2 \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top$, whose log-determinant is parameter-dependent and cannot be discarded.

Derivation. The matrix determinant lemma, $|\mathbf{A} + \mathbf{u}\mathbf{v}^\top| = |\mathbf{A}| (1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u})$, applied to $\mathbf{A} = \mathbf{V}_j$, $\mathbf{u} = \tau^2 \mathbf{1}_{n_j}$, and $\mathbf{v} = \mathbf{1}_{n_j}$, gives

$$\log |\boldsymbol{\Sigma}_j(\tau^2)| = \log |\mathbf{V}_j| + \log \left(1 + \tau^2 \mathbf{1}_{n_j}^\top \mathbf{V}_j^{-1} \mathbf{1}_{n_j} \right). \quad (\text{B.16})$$

Because $\mathbf{V}_j$ is positive definite, so is $\mathbf{V}_j^{-1}$, and therefore

$$\mathbf{1}_{n_j}^\top \mathbf{V}_j^{-1} \mathbf{1}_{n_j} > 0. \quad (\text{B.17})$$

In the equicorrelated specialization the scalar (B.17) is available in closed form: since $\mathbf{1}_{n_j}$ is the between-cluster eigenvector with $\mathbf{V}_j \mathbf{1}_{n_j} = a_j \{1 + (n_j - 1)\rho\} \mathbf{1}_{n_j}$, evaluating (B.9) at $\mathbf{1}_{n_j}$ gives

$$\begin{aligned} \mathbf{1}_{n_j}^\top \mathbf{V}_j^{-1} \mathbf{1}_{n_j} &= \frac{n_j}{a_j \{1 + (n_j - 1)\rho\}} > 0, \\ \log |\boldsymbol{\Sigma}_j(\tau^2)| &= \log |\mathbf{V}_j| + \log \left(1 + \frac{n_j \tau^2}{a_j \{1 + (n_j - 1)\rho\}} \right), \end{aligned}$$

strictly increasing in τ^2 (for a singleton cluster the same closed form gives $1/a_j$). The marginal determinant is thus part of the parameter-dependent geometry of the marginal $\boldsymbol{\xi}$ -likelihood and is not optional within either marginal likelihood. What the identity delivers is narrower and exact: because A5–A6 make $\mathbf{V}_j$ the same matrix in both interpretations, $\boldsymbol{\Sigma}_j(\tau^2)$ is identical across the two interpretations for every τ^2 , so the marginal determinant, and with it every term of the marginal log likelihood, agrees pointwise in τ^2 . The cancellation is a consequence of the matched covariance, not of the determinant being negligible; Appendix C relies on exactly this distinction when it forms the marginal fixed-effect target. $\square$

B.5. Proof of Proposition 1. Proposition 1 (restated; cluster-constant predictors are not required). Neither the clusterwise likelihood identity of Theorem 1 nor the pointwise agreement of the marginal $\boldsymbol{\xi}$ -likelihood requires special condition S1 (cluster-constant design). They require only the same observation-level design matrix $\mathbf{X}_j$ in the two interpretations, as guaranteed by A3; the rows of $\mathbf{X}_j$ may vary within cluster.

Proof. Both interpretations retain the same observation-level $\mathbf{X}_j$ and the same linear predictor $\mathbf{X}_j\boldsymbol{\beta} + \theta_j\mathbf{1}_{n_j}$, so the residual $\mathbf{r}_j(\boldsymbol{\eta})$ of (B.2) is the identical function of $\boldsymbol{\eta}$ whether or not the rows of $\mathbf{X}_j$ are constant within cluster. The conditional log likelihood (B.3) depends on the design only through $\mathbf{r}_j(\boldsymbol{\eta})$, so the clusterwise identity (B.12) holds verbatim with within-cluster covariate variation present. For the marginal statement, the marginal log likelihood under (B.5) depends on the design only through the marginal mean $\mathbf{X}_j\boldsymbol{\beta}$ and the matched covariance $\boldsymbol{\Sigma}_j(\tau^2)$; neither is altered by within-cluster covariate variation, and A5–A6 make $\boldsymbol{\Sigma}_j(\tau^2)$ the same matrix on both interpretations for every τ^2 , so the two marginal $\boldsymbol{\xi}$ -likelihoods agree pointwise. Neither interpretation replaces $\mathbf{X}_j$ by a cluster mean or any cluster-level summary, so no aggregation occurs on either side. S1 would be needed only if a later derivation replaced the observation-level design with cluster means or a closed-form within-cluster summary, which this paper does not do. $\square$

The role of S1 marks where the present identity differs in kind from aggregation-based equivalences: the equivalences of Pustejovsky and Chen (2024) operate by averaging or subgroup reclassification, so their reduced designs are necessarily study-level, whereas the identity here holds between two unaggregated interpretations of the same observation-level model and therefore covers exactly the syntheses whose moderators vary within study.

B.6. Proof of Proposition 2. Because the two log likelihoods are the same function of $\boldsymbol{\eta}$ under the matched normalization, and because any alternative normalizing term is free of $\boldsymbol{\eta}$, a sequence of consequences follows at once.

Proposition 2 (restated; immediate consequences of the identity). Under the hypotheses of Theorem 1, the following hold.

(i) *Score and Hessian.* For every $\boldsymbol{\eta}$,

$$\nabla_{\boldsymbol{\eta}} \ell^{\mathbf{M}}(\boldsymbol{\eta}) = \nabla_{\boldsymbol{\eta}} \ell^{\mathbf{S}}(\boldsymbol{\eta}), \quad \nabla_{\boldsymbol{\eta}}^2 \ell^{\mathbf{M}}(\boldsymbol{\eta}) = \nabla_{\boldsymbol{\eta}}^2 \ell^{\mathbf{S}}(\boldsymbol{\eta}). \quad (\text{B.18})$$

(ii) *Cluster scores.* Writing the likelihood-only cluster score as $\mathbf{s}_j^{\mathbf{R}}(\boldsymbol{\eta}) = \nabla_{\boldsymbol{\eta}} \ell_j^{\mathbf{R}}(\boldsymbol{\eta})$ for $\mathbf{R} \in \{\mathbf{M}, \mathbf{S}\}$, the cluster scores coincide cluster by cluster,

$$\mathbf{s}_j^{\mathbf{M}}(\boldsymbol{\eta}) = \mathbf{s}_j^{\mathbf{S}}(\boldsymbol{\eta}), \quad j = 1, \dots, J. \quad (\text{B.19})$$

(iii) *Posterior mode.* Under matched priors on the common $\boldsymbol{\zeta}$ parameterization with common support, the meta-analytic and survey posterior log densities differ by

at most a parameter-free constant. Their stationary sets and MAP sets therefore coincide, and if the posterior mode is unique, the MAP estimates are equal.

(iv) *Posterior measure.* Under matched priors, common support, propriety, and the shared parameter space of $\boldsymbol{\zeta} = (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top, \tau^2)^\top$,

$$\Pi_{\mathbf{M}}(A \mid \mathbf{y}) = \Pi_{\mathbf{S}}(A \mid \mathbf{y}) \quad (\text{B.20})$$

for every measurable set A in the common parameter space.

Proof. For parts (i) and (ii), the cluster score $\mathbf{s}_j(\boldsymbol{\eta}) = \nabla_{\boldsymbol{\eta}} \ell_j(\boldsymbol{\eta})$ has, as its only nonzero block (in the $(\boldsymbol{\beta}, \theta_j)$ coordinates),

$$\begin{pmatrix} \nabla_{\boldsymbol{\beta}} \ell_j(\boldsymbol{\eta}) \\ \partial \ell_j(\boldsymbol{\eta}) / \partial \theta_j \end{pmatrix} = \begin{pmatrix} \mathbf{X}_j^\top \mathbf{V}_j^{-1} \mathbf{r}_j(\boldsymbol{\eta}) \\ \mathbf{1}_{n_j}^\top \mathbf{V}_j^{-1} \mathbf{r}_j(\boldsymbol{\eta}) \end{pmatrix}, \quad (\text{B.21})$$

obtained from (B.3) using $\partial \mathbf{r}_j / \partial \boldsymbol{\beta} = -\mathbf{X}_j$ and $\partial \mathbf{r}_j / \partial \theta_j = -\mathbf{1}_{n_j}$; as a gradient in the full vector $\boldsymbol{\eta} = (\boldsymbol{\beta}^\top, \theta_1, \dots, \theta_J)^\top$, this cluster score has zero entries in the θ_k coordinates for $k \neq j$. The $\boldsymbol{\beta}$ -block of the total score aggregates (B.21) over clusters, while the θ_j -block is local to cluster j because ℓ_j depends on $\boldsymbol{\theta}$ only through θ_j . The corresponding nonzero local block of the negative Hessian, in the same $(\boldsymbol{\beta}, \theta_j)$ coordinates, is

$$-\nabla_{(\boldsymbol{\beta}, \theta_j)}^2 \ell_j(\boldsymbol{\eta}) = \begin{pmatrix} \mathbf{X}_j^\top \mathbf{V}_j^{-1} \mathbf{X}_j & \mathbf{X}_j^\top \mathbf{V}_j^{-1} \mathbf{1}_{n_j} \\ \mathbf{1}_{n_j}^\top \mathbf{V}_j^{-1} \mathbf{X}_j & \mathbf{1}_{n_j}^\top \mathbf{V}_j^{-1} \mathbf{1}_{n_j} \end{pmatrix}, \quad (\text{B.22})$$

with zero cross-blocks in the θ_k coordinates for $k \neq j$; this block is constant in $\boldsymbol{\eta}$ because the conditional model is linear in $\boldsymbol{\eta}$ with known $\mathbf{V}_j$. Every entry of (B.21) and (B.22) is built from $\mathbf{X}_j$, $\mathbf{1}_{n_j}$, $\mathbf{V}_j^{-1}$, and $\mathbf{r}_j(\boldsymbol{\eta})$, each shared by the two interpretations under A2–A6. Hence the cluster scores coincide, giving (B.19), and summing over clusters gives (B.18). Equivalently and without coordinates: $\ell^{\mathbf{M}}$ and $\ell^{\mathbf{S}}$ are the same function of $\boldsymbol{\eta}$ by Theorem 1, so all $\boldsymbol{\eta}$ -derivatives agree; under an alternative normalization they differ by a parameter-free constant whose gradient and Hessian vanish.

For parts (iii) and (iv), let $\pi_0(\boldsymbol{\zeta})$ denote the matched prior density on the common support, including the hierarchical contribution for $\boldsymbol{\theta} \mid \tau^2$ and any prior terms for $(\boldsymbol{\beta}, \tau^2)$. The unnormalized posterior density under interpretation R is proportional to $\exp\{\ell^{\mathbf{R}}(\boldsymbol{\eta})\} \pi_0(\boldsymbol{\zeta})$. Because $\ell^{\mathbf{M}}(\boldsymbol{\eta}) = \ell^{\mathbf{S}}(\boldsymbol{\eta})$ under the matched normalization (or differs only by a parameter-free constant absorbed into the normalizing constant), the two unnormalized posteriors are the same function on the shared parameter space, so

they share maximizers, giving (iii). Propriety guarantees finite normalizing constants, and for any measurable set A ,

$$\Pi_M(A \mid \mathbf{y}) = \frac{\int_A \exp\{\ell^M(\boldsymbol{\eta})\} \pi_0(\boldsymbol{\zeta}) d\boldsymbol{\zeta}}{\int \exp\{\ell^M(\boldsymbol{\eta})\} \pi_0(\boldsymbol{\zeta}) d\boldsymbol{\zeta}} = \frac{\int_A \exp\{\ell^S(\boldsymbol{\eta})\} \pi_0(\boldsymbol{\zeta}) d\boldsymbol{\zeta}}{\int \exp\{\ell^S(\boldsymbol{\eta})\} \pi_0(\boldsymbol{\zeta}) d\boldsymbol{\zeta}} = \Pi_S(A \mid \mathbf{y}), \quad (\text{B.23})$$

which is (B.20). $\square$

Two clarifications fix the scope of Proposition 2. First, the derivative identities (B.18)–(B.19) are conditional latent-effect statements in $\boldsymbol{\eta}$: the between-cluster variance τ^2 does not enter the conditional likelihood (B.3) and appears nowhere in (B.21) or (B.22), so a covariance object built from $\boldsymbol{\eta}$ is a conditional latent-effect object, not a full-joint sandwich; the posterior-measure equality (B.20) is stated in the full hierarchical vector $\boldsymbol{\zeta}$, because τ^2 belongs to the posterior through the Level-2 distribution of θ_j ; keeping the three vectors of (B.4) distinct is what licenses the later development. Second, posterior equality is equality of probability measures: the two interpretations assign the same probability to every measurable event, and they therefore share the same posterior center whenever that center is a finite, unique functional of the posterior measure. Sampler behavior, seeds, iteration counts, and Monte Carlo error lie wholly outside the identity; the cluster-additive term that an alternative normalization would introduce is free of $\boldsymbol{\zeta}$ and factors out of Bayes’ rule.

Appendix C. The marginal fixed-effect target and the selected-block calibration

This appendix supplies the derivations that Section 4 states in compact form: the rank obstruction that closes the conditional route (Lemma 1), the marginal model and its sandwich objects, the marginal fixed-effect sandwich identity (Theorem 2), the selected reported block, and the center-preserving calibration (Corollary 1). It builds on the likelihood identity of Appendix B. Displays carry appendix-local numbers (C.k); cross-document pointers are written in prose. The three parameter vectors are kept strictly distinct throughout,

$$\boldsymbol{\eta} = (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top)^\top, \quad \boldsymbol{\zeta} = (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top, \tau^2)^\top, \quad \boldsymbol{\xi} = (\boldsymbol{\beta}^\top, \tau^2)^\top, \quad (\text{C.1})$$

with $\boldsymbol{\theta} = (\theta_1, \dots, \theta_J)^\top$. The target of this appendix is defined in $\boldsymbol{\xi}$, and it is reported for the block $\boldsymbol{\psi} = \mathbf{R}_\psi \boldsymbol{\beta}$, which is not the same object as $\boldsymbol{\xi}$.

C.1. Why the conditional latent-effect sandwich is not the target. Appendix B proved that the likelihood-only conditional cluster scores coincide across the two interpretations, so the conditional latent-effect sandwich is a faithful object on which the interpretations agree. It is nonetheless the wrong object to report for moderator inference. The target is marginal because the reported quantity integrates over the latent cluster effects $\boldsymbol{\theta}$ (the estimand, per Proposition 2 of Appendix B); separately, Lemma 1 establishes that the conditional latent-effect meat is rank-deficient in the full joint vector $\boldsymbol{\eta}$, which rules out full-joint Cholesky calibration on rank grounds alone. Let $\hat{\boldsymbol{\eta}}$ be a common conditional evaluation point and let $\mathbf{s}_j(\hat{\boldsymbol{\eta}}) = \nabla_{\boldsymbol{\eta}} \ell_j(\hat{\boldsymbol{\eta}})$ be the likelihood-only conditional cluster score of cluster j . The conditional latent-effect meat is the outer-product sum

$$\mathbf{M}_{\boldsymbol{\eta}} = \sum_{j=1}^J \mathbf{s}_j(\hat{\boldsymbol{\eta}}) \mathbf{s}_j(\hat{\boldsymbol{\eta}})^\top. \quad (\text{C.2})$$

Lemma 1 (restated; rank limitation of the conditional latent-effect meat). With $\boldsymbol{\eta} = (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top)^\top$ and likelihood-only cluster scores $\mathbf{s}_j(\hat{\boldsymbol{\eta}}) = \nabla_{\boldsymbol{\eta}} \ell_j(\hat{\boldsymbol{\eta}})$, the conditional latent-effect meat (C.2) satisfies

$$\text{rank}(\mathbf{M}_{\boldsymbol{\eta}}) \leq J. \quad (\text{C.3})$$

Proof. Each summand $\mathbf{s}_j(\hat{\boldsymbol{\eta}}) \mathbf{s}_j(\hat{\boldsymbol{\eta}})^\top$ is an outer product of a single vector with itself and therefore has rank at most one. A sum of J matrices each of rank at most one has rank at most J , which is (C.3). Since $\boldsymbol{\beta} \in \mathbb{R}^p$ and $\boldsymbol{\theta} \in \mathbb{R}^J$, the conditional vector has dimension $\dim(\boldsymbol{\eta}) = p + J$, so $\text{rank}(\mathbf{M}_{\boldsymbol{\eta}}) \leq J < p + J$ for every $p \geq 1$, and $\mathbf{M}_{\boldsymbol{\eta}}$ cannot have full rank. If a nonsingular bread inverse is used to form the conditional sandwich, pre- and post-multiplication by that fixed nonsingular matrix cannot increase rank, so the conditional sandwich inherits the rank deficiency of its meat. $\square$

The structural reason is that the per-cluster latent effects proliferate with the number of clusters while each cluster contributes only one score vector, so the conditional meat cannot fill out the full conditional dimension once any fixed effect is present; this is the meta-analytic instance of the incidental-parameters phenomenon (Lancaster, 2000; Neyman & Scott, 1948). Lemma 1 rules out one candidate target; it does not make conditional objects useless, because the conditional sandwich remains a genuine cross-interpretation check and is treated as such in Appendix D. The remedy is to integrate out the latent cluster effects and work in the marginal vector $\boldsymbol{\xi} = (\boldsymbol{\beta}, \tau^2)$,

of dimension $p + 1$ rather than $p + J$, carrying τ^2 explicitly as a nuisance coordinate rather than fixing it silently.

C.2. The marginal clustered Gaussian model. Integrating the latent cluster effects out of the shared conditional object returns the marginal clustered Gaussian model,

$$\mathbf{y}_j \mid \boldsymbol{\beta}, \tau^2 \sim \mathbf{N}_{n_j}(\mathbf{X}_j \boldsymbol{\beta}, \boldsymbol{\Sigma}_j(\tau^2)), \quad \boldsymbol{\Sigma}_j(\tau^2) = \mathbf{V}_j(a_j, \rho) + \tau^2 \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top, \quad (\text{C.4})$$

with marginal parameter $\boldsymbol{\xi} = (\boldsymbol{\beta}^\top, \tau^2)^\top$ and $\mathbf{V}_j(a_j, \rho) = a_j \mathbf{C}_j(\rho)$ the matched known within-cluster covariance. Writing the marginal residual

$$\mathbf{u}_j(\boldsymbol{\beta}) = \mathbf{y}_j - \mathbf{X}_j \boldsymbol{\beta}, \quad (\text{C.5})$$

the marginal cluster log likelihood is

$$\ell_j^{\text{marg}}(\boldsymbol{\xi}) = -\frac{1}{2} \{n_j \log(2\pi) + \log |\boldsymbol{\Sigma}_j(\tau^2)| + \mathbf{u}_j(\boldsymbol{\beta})^\top \boldsymbol{\Sigma}_j(\tau^2)^{-1} \mathbf{u}_j(\boldsymbol{\beta})\}, \quad (\text{C.6})$$

with total $\ell^{\text{marg}}(\boldsymbol{\xi}) = \sum_{j=1}^J \ell_j^{\text{marg}}(\boldsymbol{\xi})$. The determinant $\log |\boldsymbol{\Sigma}_j(\tau^2)|$ is retained: Appendix B (Caveat 2) proved it parameter-dependent in τ^2 , so it is not a removable constant, and it agrees across the two interpretations because they share the same $\boldsymbol{\Sigma}_j(\tau^2)$ under $\mathcal{T}_A$, not because it can be discarded. The marginal model uses the observation-level $\mathbf{X}_j$, so within-cluster moderator variation is preserved; by Proposition 1 of Appendix B, special condition S1 is not required.

C.3. Marginal scores, bread, meat, and sandwich. The marginal cluster score is $\mathbf{s}_j^{\text{marg}}(\boldsymbol{\xi}) = \nabla_{\boldsymbol{\xi}} \ell_j^{\text{marg}}(\boldsymbol{\xi})$. Because $\boldsymbol{\Sigma}_j(\tau^2)$ does not depend on $\boldsymbol{\beta}$ and $\partial \boldsymbol{\Sigma}_j(\tau^2) / \partial \tau^2 = \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top$, differentiating (C.6) gives the two blocks

$$\mathbf{s}_j^{\text{marg}}(\boldsymbol{\xi}) = \begin{pmatrix} \mathbf{X}_j^\top \boldsymbol{\Sigma}_j(\tau^2)^{-1} \mathbf{u}_j(\boldsymbol{\beta}) \\ -\frac{1}{2} \left[\mathbf{1}_{n_j}^\top \boldsymbol{\Sigma}_j(\tau^2)^{-1} \mathbf{1}_{n_j} - \left(\mathbf{1}_{n_j}^\top \boldsymbol{\Sigma}_j(\tau^2)^{-1} \mathbf{u}_j(\boldsymbol{\beta}) \right)^2 \right] \end{pmatrix}. \quad (\text{C.7})$$

The first block is the fixed-effect score, from $\partial \mathbf{u}_j / \partial \boldsymbol{\beta} = -\mathbf{X}_j$. The second block is the scalar τ^2 score: differentiating the log determinant gives $\partial \log |\boldsymbol{\Sigma}_j| / \partial \tau^2 = \text{tr}(\boldsymbol{\Sigma}_j^{-1} \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top)$, which equals $\mathbf{1}_{n_j}^\top \boldsymbol{\Sigma}_j^{-1} \mathbf{1}_{n_j}$, and differentiating the quadratic form gives $\partial (\mathbf{u}_j^\top \boldsymbol{\Sigma}_j^{-1} \mathbf{u}_j) / \partial \tau^2 = -(\mathbf{1}_{n_j}^\top \boldsymbol{\Sigma}_j^{-1} \mathbf{u}_j)^2$ through $\partial \boldsymbol{\Sigma}_j^{-1} / \partial \tau^2 = -\boldsymbol{\Sigma}_j^{-1} \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top \boldsymbol{\Sigma}_j^{-1}$. Equation

(C.7) makes explicit that τ^2 enters the marginal score and is carried as a nuisance coordinate of $\boldsymbol{\xi}$, not silently fixed.

Fix the evaluation point $\hat{\boldsymbol{\xi}}$ at an interior sample root where $\sum_j \mathbf{s}_j^{\text{marg}}(\hat{\boldsymbol{\xi}}) = 0$ and $\mathbf{H}_{\boldsymbol{\xi}}$ is positive definite; let $\boldsymbol{\xi}^\dagger$ denote the corresponding population pseudo-true root. The affine-recoloring fixed point, that is, the reported center, is the posterior mean $\bar{\boldsymbol{\xi}}_{\text{post}}$ of Proposition 2 of Appendix B, which under the stated regularity conditions is consistent for $\boldsymbol{\xi}^\dagger$. The likelihood-only marginal meat is the outer-product sum of marginal cluster scores,

$$\mathbf{M}_{\boldsymbol{\xi}} = \sum_{j=1}^J \mathbf{s}_j^{\text{marg}}(\hat{\boldsymbol{\xi}}) \mathbf{s}_j^{\text{marg}}(\hat{\boldsymbol{\xi}})^\top, \quad (\text{C.8})$$

and, under a named bread convention, the bread is the likelihood-only negative marginal Hessian

$$\mathbf{H}_{\boldsymbol{\xi}} = -\nabla_{\boldsymbol{\xi}}^2 \ell^{\text{marg}}(\hat{\boldsymbol{\xi}}), \quad (\text{C.9})$$

taken likelihood-only, with no prior curvature; a posterior-Hessian bread that adds prior curvature is a separate named convention, not the reported target. When $\mathbf{H}_{\boldsymbol{\xi}}$ is nonsingular, the marginal fixed-effect sandwich is

$$\mathbf{V}_{\boldsymbol{\xi}}^{\text{marg}} = \mathbf{H}_{\boldsymbol{\xi}}^{-1} \mathbf{M}_{\boldsymbol{\xi}} \mathbf{H}_{\boldsymbol{\xi}}^{-1}. \quad (\text{C.10})$$

One convention is fixed and used throughout: the meat (C.8) is built from likelihood-only marginal cluster scores; prior curvature may enter the bread under the named posterior-bread convention, but prior scores never enter the meat.

The rank behavior of $\mathbf{M}_{\boldsymbol{\xi}}$ deserves a careful statement. Because $\mathbf{M}_{\boldsymbol{\xi}}$ is a $(p+1) \times (p+1)$ matrix, the structural obstruction of Lemma 1 (one latent effect per cluster) no longer follows from the dimension count; this removes the obstruction without by itself guaranteeing full rank. At an exact marginal maximizer the cluster scores sum to the zero vector, so the outer-product meat (C.8) has rank at most $J-1$; with $p+1 \ll J$ that bound does not bind the $(p+1)$ -dimensional target, and full-rank use requires the usual score-variation, estimability, and nonsingular-bread conditions. Positive definiteness of the selected $\boldsymbol{\beta}$ -block target in Corollary 1 is accordingly a hypothesis checked at $\hat{\boldsymbol{\xi}}$, by the minimum eigenvalue, condition number, or successful Cholesky factorization of both the posterior and target covariances; the empirical designs of Appendix F (Clinton, $J = 20$; Roberts, $J = 145$) satisfy it with margin. Cluster-robust sandwich variances for clustered fixed-effect inference are standard (Cameron

& Miller, 2015), and the convention dependence is the ordinary one: changing the bread convention changes the target, and once a convention is fixed and applied identically on both interpretations the target is shared. The object transported here is the model-based marginal fixed-effect sandwich carrying heterogeneity as a nuisance coordinate; it is not a generic working-independence sandwich of the generalized-estimating-equations tradition (Liang & Zeger, 1986).

C.4. Full proof of Theorem 2. Theorem 2 (restated; marginal fixed-effect sandwich identity). Let Assumption 1 hold and let the two interpretations be identified through the restricted translation map $\mathcal{T}_A$. Suppose the same random-effect distribution $\theta_j \mid \tau^2 \sim \mathbf{N}(0, \tau^2)$ is used, the same observation-level $\mathbf{X}_j$ is retained, the same marginal evaluation point $\hat{\boldsymbol{\xi}}$ is used, and the same marginal bread/meat convention is applied on both interpretations. Then the two interpretations have identical marginal log likelihoods in $\boldsymbol{\xi}$,

$$\ell_{\mathbf{M}}^{\text{marg}}(\boldsymbol{\xi}) = \ell_{\mathbf{S}}^{\text{marg}}(\boldsymbol{\xi}) \quad \text{for every } \boldsymbol{\xi}, \quad (\text{C.11})$$

and consequently the marginal scores, bread, and meat coincide,

$$\mathbf{s}_{j,\mathbf{M}}^{\text{marg}}(\boldsymbol{\xi}) = \mathbf{s}_{j,\mathbf{S}}^{\text{marg}}(\boldsymbol{\xi}), \quad \mathbf{H}_{\boldsymbol{\xi},\mathbf{M}} = \mathbf{H}_{\boldsymbol{\xi},\mathbf{S}}, \quad \mathbf{M}_{\boldsymbol{\xi},\mathbf{M}} = \mathbf{M}_{\boldsymbol{\xi},\mathbf{S}}, \quad (\text{C.12})$$

so that, when the common bread is nonsingular, the marginal fixed-effect sandwiches are equal,

$$\mathbf{V}_{\boldsymbol{\xi},\mathbf{M}}^{\text{marg}} = \mathbf{V}_{\boldsymbol{\xi},\mathbf{S}}^{\text{marg}}. \quad (\text{C.13})$$

Proof. By Theorem 1 of Appendix B, the two interpretations share the same conditional Gaussian object $\mathbf{y}_j \mid \boldsymbol{\beta}, \theta_j \sim \mathbf{N}_{n_j}(\mathbf{X}_j \boldsymbol{\beta} + \theta_j \mathbf{1}_{n_j}, \mathbf{V}_j)$ with the same known $\mathbf{V}_j$. Integrating this common object over the common random-effect distribution $\theta_j \mid \tau^2 \sim \mathbf{N}(0, \tau^2)$ gives the same marginal density on the two interpretations:

$$\begin{aligned} p_{\mathbf{M}}(\mathbf{y}_j \mid \boldsymbol{\xi}) &= \int p_{\mathbf{M}}(\mathbf{y}_j \mid \boldsymbol{\beta}, \theta_j) p(\theta_j \mid \tau^2) d\theta_j \\ &= \int p_{\mathbf{S}}(\mathbf{y}_j \mid \boldsymbol{\beta}, \theta_j) p(\theta_j \mid \tau^2) d\theta_j \\ &= p_{\mathbf{S}}(\mathbf{y}_j \mid \boldsymbol{\xi}), \end{aligned} \quad (\text{C.14})$$

the middle equality holding because the two integrands are equal pointwise in θ_j by Theorem 1 and the integrating measure is the same. The common integral is the Gaussian density of the marginal model (C.4); equivalently, conditions A5–A6 make

$\Sigma_j(\tau^2)$ the same matrix on both interpretations for every τ^2 , so the marginal determinant $\log |\Sigma_j(\tau^2)|$, the inverse covariance $\Sigma_j(\tau^2)^{-1}$, the marginal residual $\mathbf{u}_j(\boldsymbol{\beta})$, and the quadratic form in (C.6) agree term by term. The determinant is retained throughout and cancels across interpretations because it is the same determinant on both sides, not because it is parameter-free. Summing the equal cluster contributions over the independent clusters gives (C.11). Differentiating the same function ℓ^{marg} with respect to the same $\boldsymbol{\xi}$ and evaluating at the same point $\hat{\boldsymbol{\xi}}$ gives the same marginal cluster scores (C.7), the same negative Hessian (C.9), and the same outer-product meat (C.8), which is (C.12); assembling these by (C.10) gives (C.13). The argument retains the observation-level $\mathbf{X}_j$, does not invoke special condition S1, and carries τ^2 as a nuisance coordinate of $\boldsymbol{\xi}$ rather than fixing it. $\square$

Theorem 2 is a narrowly scoped translation result: it identifies when a survey-side covariance-calibration target is, algebraically, the target induced by a Bayesian correlated-effects meta-analysis, which is the payoff the calibrated-Bayes program attaches to a well-chosen named object (Little, 2012; Rubin, 1984; M. R. Williams & Savitsky, 2021). Where Pustejovsky and Chen (2024) relate aggregated or subgrouped analyses to multivariate models, Theorem 2 identifies, on the unaggregated boundary, a fixed-effect covariance target the Bayesian analyst can adopt without leaving the meta-analytic model or changing the unit of analysis.

C.5. The reported block and the selected-block target. Analysts rarely report the full $\boldsymbol{\xi}$ vector; they report a fixed-effect block or contrast. Let the reported quantity be

$$\boldsymbol{\psi} = \mathbf{R}_\psi \boldsymbol{\beta}, \quad (\text{C.15})$$

where $\mathbf{R}_\psi$ is a fixed $q \times p$ selection or contrast matrix acting on the fixed effects. The target covariance for $\boldsymbol{\psi}$ is the corresponding block of the marginal sandwich,

$$\mathbf{V}_\psi^{\text{target}} = \mathbf{R}_\psi \mathbf{V}_\xi^{\text{marg}}[\boldsymbol{\beta}, \boldsymbol{\beta}] \mathbf{R}_\psi^\top, \quad (\text{C.16})$$

where $\mathbf{V}_\xi^{\text{marg}}[\boldsymbol{\beta}, \boldsymbol{\beta}]$ is the $\boldsymbol{\beta}$ – $\boldsymbol{\beta}$ submatrix of (C.10). Because (C.10) is formed in the nuisance vector $\boldsymbol{\xi} = (\boldsymbol{\beta}, \tau^2)$, the selected block already reflects the uncertainty contributed by τ^2 through the marginal score (C.7); it is not a $\boldsymbol{\beta}$ -only sandwich computed at a fixed τ^2 , not the conditional $\boldsymbol{\eta}$ -block ruled out by Lemma 1, and not a small-sample-corrected object by default.

The posterior object to be calibrated is the posterior covariance of the same reported block,

$$\Sigma_{\text{post},\psi} = \text{Var}_{\Pi}(\boldsymbol{\psi} \mid \mathbf{y}), \quad (\text{C.17})$$

where Π is the common posterior measure of Proposition 2 of Appendix B. When the posterior mean exists, the shared posterior center is

$$\bar{\boldsymbol{\psi}}_{\text{post}} = \mathbb{E}_{\Pi}(\boldsymbol{\psi} \mid \mathbf{y}) = \mathbf{R}_{\psi} \mathbb{E}_{\Pi}(\boldsymbol{\beta} \mid \mathbf{y}). \quad (\text{C.18})$$

Because Proposition 2 gives equality of posterior measures under matched priors, this center is the same under the two interpretations, which is what makes the selected-block calibration a cross-interpretation consequence rather than a merely procedural adjustment. As implementation provenance only: the `bayesRVE` package reads its selected-block target off the marginal $\boldsymbol{\xi}$ -sandwich, never off a conditional or latent object, which is the choice Lemma 1 motivates and Theorem 2 justifies (Lee, 2026).

C.6. Selected-block covariance calibration. The calibration step uses an affine whiten-then-recolor map to preserve the shared center while replacing the posterior covariance block with the marginal sandwich target.

Corollary 1 (restated; selected-block covariance-calibration consequence). Let $\boldsymbol{\psi} = \mathbf{R}_{\psi}\boldsymbol{\beta}$ with target $\mathbf{V}_{\psi}^{\text{target}}$ given by (C.16), and suppose $\Sigma_{\text{post},\psi}$ and $\mathbf{V}_{\psi}^{\text{target}}$ are positive definite, with Cholesky factors

$$\Sigma_{\text{post},\psi} = \mathbf{L}_{\text{post}}^{\psi} \mathbf{L}_{\text{post}}^{\psi\top}, \quad \mathbf{V}_{\psi}^{\text{target}} = \mathbf{L}_{\text{target}}^{\psi} \mathbf{L}_{\text{target}}^{\psi\top}. \quad (\text{C.19})$$

Then the affine covariance-calibration map

$$\mathcal{C}_{\psi}(\boldsymbol{\psi}) = \bar{\boldsymbol{\psi}}_{\text{post}} + \mathbf{L}_{\text{target}}^{\psi} \left(\mathbf{L}_{\text{post}}^{\psi} \right)^{-1} (\boldsymbol{\psi} - \bar{\boldsymbol{\psi}}_{\text{post}}) \quad (\text{C.20})$$

preserves the shared center,

$$\mathbb{E}\{\mathcal{C}_{\psi}(\boldsymbol{\psi}) \mid \mathbf{y}\} = \bar{\boldsymbol{\psi}}_{\text{post}}, \quad (\text{C.21})$$

and has covariance equal to the target,

$$\begin{aligned}
\text{Var}\{\mathcal{C}_\psi(\psi) \mid \mathbf{y}\} &= \mathbf{L}_{\text{target}}^\psi \left(\mathbf{L}_{\text{post}}^\psi \right)^{-1} \Sigma_{\text{post},\psi} \left(\mathbf{L}_{\text{post}}^\psi \right)^{-\top} \mathbf{L}_{\text{target}}^{\psi\top} \\
&= \mathbf{L}_{\text{target}}^\psi \mathbf{L}_{\text{target}}^{\psi\top} \\
&= \mathbf{V}_\psi^{\text{target}}.
\end{aligned} \tag{C.22}$$

Proof. The map (C.20) is affine in ψ with fixed point $\bar{\psi}_{\text{post}}$. Because $\bar{\psi}_{\text{post}}$ is the posterior mean, $E(\psi - \bar{\psi}_{\text{post}} \mid \mathbf{y}) = \mathbf{0}$, so taking the conditional expectation of (C.20) gives $E\{\mathcal{C}_\psi(\psi) \mid \mathbf{y}\} = \bar{\psi}_{\text{post}} + \mathbf{L}_{\text{target}}^\psi (\mathbf{L}_{\text{post}}^\psi)^{-1} \mathbf{0} = \bar{\psi}_{\text{post}}$, which is (C.21). For the covariance, $\text{Var}\{\mathcal{C}_\psi(\psi) \mid \mathbf{y}\} = \mathbf{A} \Sigma_{\text{post},\psi} \mathbf{A}^\top$ with $\mathbf{A} = \mathbf{L}_{\text{target}}^\psi (\mathbf{L}_{\text{post}}^\psi)^{-1}$. Substituting $\Sigma_{\text{post},\psi} = \mathbf{L}_{\text{post}}^\psi \mathbf{L}_{\text{post}}^{\psi\top}$ and using $(\mathbf{L}_{\text{post}}^\psi)^{-1} \mathbf{L}_{\text{post}}^\psi = \mathbf{I}$ and $\mathbf{L}_{\text{post}}^{\psi\top} (\mathbf{L}_{\text{post}}^\psi)^{-\top} = \mathbf{I}$ cancels the posterior factors and leaves $\mathbf{L}_{\text{target}}^\psi \mathbf{L}_{\text{target}}^{\psi\top} = \mathbf{V}_\psi^{\text{target}}$, which is (C.22). Positive definiteness guarantees that the Cholesky factors and $(\mathbf{L}_{\text{post}}^\psi)^{-1}$ exist; Proposition 2 of Appendix B guarantees that $\bar{\psi}_{\text{post}}$ is shared across interpretations; and Theorem 2 guarantees that $\mathbf{V}_\psi^{\text{target}}$ is identical across interpretations once the β -block is selected and $\mathbf{R}_\psi$ applied, provided the same marginal bread/meat convention is used. $\square$

Three guards delimit Corollary 1 and are part of it. Positive definiteness is substantive: if either $\Sigma_{\text{post},\psi}$ or $\mathbf{V}_\psi^{\text{target}}$ is singular, the map (C.20) is not defined as stated, and generalized-inverse or low-rank variants would be separate extensions not developed here. The map is a covariance alignment, not a coverage theorem: it fixes which covariance object is reported and says nothing about the frequentist coverage of the resulting intervals, which would require additional assumptions (Bell & McCaffrey, 2002; Imbens & Kolesár, 2016). And the target is the selected β -block only; calibrating the θ , τ^2 , or full- ζ blocks, or adopting a profile or small-sample-corrected target, would each require its own target theory before being claimed. The result inherits the scope of the bridge: the matched homoscedastic equicorrelated specialization $\mathbf{V}_j = a_j \mathbf{C}_j(\rho)$, with the cluster-robust marginal sandwich providing the reported fixed-effect covariance target.

Appendix D. Matched-convention supporting results: the conditional sandwich and aligned HC/CR corrections

This appendix records the supporting machinery that surrounds the marginal target without displacing it: the conditional latent-effect sandwich equality (Proposition 3), the bread conventions that equality requires when the conditional information is singular, and the aligned heteroscedasticity-consistent and cluster-robust (HC/CR) correction equality (Corollary 2). Neither the conditional sandwich nor any HC/CR-corrected object is the reported fixed-effect target; they are cross-interpretation checks

on the bridge and a record of which correction ingredients must be matched. Displays carry appendix-local numbers (D.k); cross-document pointers are written in prose. The parameter vectors are

$$\boldsymbol{\eta} = (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top)^\top, \quad \boldsymbol{\zeta} = (\boldsymbol{\beta}^\top, \boldsymbol{\theta}^\top, \tau^2)^\top, \quad \boldsymbol{\xi} = (\boldsymbol{\beta}^\top, \tau^2)^\top, \quad (\text{D.1})$$

with $\boldsymbol{\theta} = (\theta_1, \dots, \theta_J)^\top$, and the proofs of this appendix live in the conditional vector $\boldsymbol{\eta}$. The reported quantity and its target remain the Appendix C objects,

$$\boldsymbol{\psi} = \mathbf{R}_\psi \boldsymbol{\beta}, \quad \mathbf{V}_\psi^{\text{target}} = \mathbf{R}_\psi \mathbf{V}_\xi^{\text{marg}} [\boldsymbol{\beta}, \boldsymbol{\beta}] \mathbf{R}_\psi^\top, \quad (\text{D.2})$$

which this appendix neither redefines nor displaces.

D.1. Conditional latent-effect objects and the bread/meat convention. The conditional latent-effect sandwich is built in $\boldsymbol{\eta} = (\boldsymbol{\beta}, \boldsymbol{\theta})$ from the same conditional Gaussian object the bridge uses. Recall the cluster residual and the clusterwise conditional Gaussian log likelihood,

$$\mathbf{r}_j(\boldsymbol{\eta}) = \mathbf{y}_j - \mathbf{X}_j \boldsymbol{\beta} - \theta_j \mathbf{1}_{n_j}, \quad (\text{D.3})$$

$$\ell_j(\boldsymbol{\eta}) = -\frac{1}{2} \left\{ n_j \log(2\pi) + \log |\mathbf{V}_j| + \mathbf{r}_j(\boldsymbol{\eta})^\top \mathbf{V}_j^{-1} \mathbf{r}_j(\boldsymbol{\eta}) \right\}, \quad (\text{D.4})$$

with total $\ell(\boldsymbol{\eta}) = \sum_{j=1}^J \ell_j(\boldsymbol{\eta})$. Under $\mathcal{T}_A$, Appendix B (Theorem 1 and Proposition 2) shows that the two interpretations have the same conditional likelihood, the same likelihood-only cluster scores $\mathbf{s}_j(\boldsymbol{\eta}) = \nabla_{\boldsymbol{\eta}} \ell_j(\boldsymbol{\eta})$, and the same likelihood Hessian at every common $\boldsymbol{\eta}$; the score $\mathbf{s}_j(\boldsymbol{\eta})$ sits in the full vector with zero entries in the θ_k coordinates for $k \neq j$. Evaluated at a common conditional point $\hat{\boldsymbol{\eta}}$, the conditional latent-effect meat is the outer-product sum

$$\mathbf{M}_\eta = \sum_{j=1}^J \mathbf{s}_j(\hat{\boldsymbol{\eta}}) \mathbf{s}_j(\hat{\boldsymbol{\eta}})^\top, \quad \text{rank}(\mathbf{M}_\eta) \leq J < p + J \quad (p \geq 1). \quad (\text{D.5})$$

The rank bound is Lemma 1 of Appendix C, recalled rather than re-proved: the conditional meat cannot supply a full-rank covariance target for arbitrary reported fixed-effect contrasts, which is why the conditional sandwich below is a cross-interpretation check and not a competitor to the marginal target.

Forming a sandwich also requires a bread, and because the conditional parameterization carries a known singularity (Section D.3), the bread convention must be

named. Throughout this appendix the likelihood bread, the prior curvature, and the posterior bread are

$$\mathbf{H}_\ell = -\nabla_{\boldsymbol{\eta}}^2 \ell(\hat{\boldsymbol{\eta}}), \quad \mathbf{H}_\pi = -\nabla_{\boldsymbol{\eta}}^2 \log \pi(\hat{\boldsymbol{\eta}}), \quad \mathbf{H}_{\text{post}} = \mathbf{H}_\ell + \mathbf{H}_\pi, \quad (\text{D.6})$$

and $\mathbf{H}_\eta$ denotes whichever named bread convention (likelihood-only, posterior, constrained, or estimable-subspace) is in force. One rule is fixed throughout: the meat is built from likelihood-only cluster scores; prior curvature may enter the bread under the posterior-Hessian convention, but prior scores never enter the meat; and the same named convention must be applied on both interpretations.

D.2. Full proof of Proposition 3. Proposition 3 (restated; conditional latent-effect sandwich equality). Under the hypotheses of Theorem 1 and Proposition 2, with matched priors, if the two interpretations use the same evaluation point $\hat{\boldsymbol{\eta}}$ and the same bread convention, then the conditional meats and breads coincide,

$$\mathbf{M}_{\eta, \text{M}} = \mathbf{M}_{\eta, \text{S}}, \quad \mathbf{H}_{\eta, \text{M}} = \mathbf{H}_{\eta, \text{S}}, \quad (\text{D.7})$$

and therefore, when the chosen bread inverse is well-defined, the conditional latent-effect sandwiches coincide,

$$\mathbf{V}_{\eta, \text{M}}^{\text{cond}} = \mathbf{H}_{\eta, \text{M}}^{-1} \mathbf{M}_{\eta, \text{M}} \mathbf{H}_{\eta, \text{M}}^{-1} = \mathbf{H}_{\eta, \text{S}}^{-1} \mathbf{M}_{\eta, \text{S}} \mathbf{H}_{\eta, \text{S}}^{-1} = \mathbf{V}_{\eta, \text{S}}^{\text{cond}}. \quad (\text{D.8})$$

Proof. By Theorem 1 of Appendix B the two interpretations have the same conditional log likelihood as a function of $\boldsymbol{\eta}$. Differentiating that common function, Proposition 2 gives identical likelihood-only cluster scores and an identical likelihood Hessian at any common evaluation point: $\mathbf{s}_{j, \text{M}}(\hat{\boldsymbol{\eta}}) = \mathbf{s}_{j, \text{S}}(\hat{\boldsymbol{\eta}})$ for every j , and $-\nabla_{\boldsymbol{\eta}}^2 \ell_{\text{M}}(\hat{\boldsymbol{\eta}}) = -\nabla_{\boldsymbol{\eta}}^2 \ell_{\text{S}}(\hat{\boldsymbol{\eta}})$. Substituting the equal cluster scores into (D.5) gives $\mathbf{M}_{\eta, \text{M}} = \mathbf{M}_{\eta, \text{S}}$, and the equal Hessian gives the equal likelihood bread $\mathbf{H}_\ell$. With matched priors the prior curvature $\mathbf{H}_\pi$ coincides as well, so the posterior bread $\mathbf{H}_{\text{post}} = \mathbf{H}_\ell + \mathbf{H}_\pi$ coincides; the constrained and estimable-subspace breads of Section D.3 are deterministic functions of these same shared ingredients, so whichever named bread $\mathbf{H}_\eta$ is in force coincides across interpretations. This gives (D.7). When the chosen $\mathbf{H}_\eta$ is invertible, or when the sandwich is formed on an explicitly stated estimable subspace on which the bread is invertible, substituting the matched meat and bread into $\mathbf{V}_\eta^{\text{cond}} = \mathbf{H}_\eta^{-1} \mathbf{M}_\eta \mathbf{H}_\eta^{-1}$ gives (D.8). $\square$

The scope of (D.7)–(D.8) is limited in three ways. The object is conditional in $\boldsymbol{\eta}$: it excludes the heterogeneity parameter τ^2 and is therefore a conditional latent-effect

object, not a full-joint sandwich; its meat is rank-limited by (D.5); and Proposition 3 is an equality conditional on a named common bread convention and a well-defined inverse, or on an explicitly stated estimable subspace when the conditional information is singular, which is the case taken up next.

D.3. Singular likelihood bread and admissible conventions. The qualifier “when the chosen bread inverse is well-defined” is not a formality, because the likelihood-only bread $\mathbf{H}_\ell$ is generically singular in the conditional parameterization. As recorded in Appendix A, a global intercept together with free cluster effects leaves a flat direction: writing β_0 for the intercept whose design column is $\mathbf{1}_{n_j}$, the reparameterization

$$\beta_0 \mapsto \beta_0 + c, \quad \theta_j \mapsto \theta_j - c \quad (j = 1, \dots, J), \quad (\text{D.9})$$

leaves every fitted conditional mean unchanged,

$$(\beta_0 + c)\mathbf{1}_{n_j} + (\theta_j - c)\mathbf{1}_{n_j} = (\beta_0 + \theta_j)\mathbf{1}_{n_j}, \quad (\text{D.10})$$

so the residuals and the known working covariances are unchanged, the conditional likelihood is flat along this direction, and $\mathbf{H}_\ell$ is not invertible. This is the classical estimability situation in which a less-than-full-rank design renders the information matrix singular, so that only estimable linear combinations are recovered, through a generalized inverse or an identifying constraint (Searle, 1971).

The two defects of the conditional object are distinct and must not be conflated: the intercept aliasing (D.9)–(D.10) concerns invertibility of the *bread*, while the rank bound (D.5) concerns the rank of the *meat*; a remedy for one does not address the other, and supplying curvature along the flat direction leaves the meat rank bounded by J . The remedy for the bread singularity is to name an admissible convention and apply it identically on both interpretations. Four are admissible.

- **Posterior-bread convention.** Under a proper prior, the prior curvature $\mathbf{H}_\pi$ supplies curvature along the flat direction, so $\mathbf{H}_{\text{post}} = \mathbf{H}_\ell + \mathbf{H}_\pi$ is nonsingular.
- **Identifying-constraint convention.** A constraint such as $\sum_j \theta_j = 0$, or a fixed intercept, removes the flat direction before the information matrix is formed.
- **Constrained-bread convention.** The bread is formed under an explicitly stated restriction matrix, with the retained coordinates made clear.

- **Estimable-subspace convention.** The sandwich is formed only on the estimable linear combinations of $\boldsymbol{\eta}$, with the inverse taken on that subspace.

Each convention is a deterministic function of the shared likelihood and prior, so Proposition 3 holds verbatim once the convention is named and applied identically on both interpretations. What is not admissible is an unnamed inversion of a singular $\mathbf{H}_\ell$, or a convention that differs between the interpretations. These are bread-invertibility devices for the conditional object, not the center-preserving calibration of Appendix C, which acts on a positive-definite reported block. The marginal $\boldsymbol{\xi}$ sandwich avoids the singularity entirely (it carries no θ_j and has dimension $p+1$), which is one more reason the marginal object, not the conditional one, is the reported target.

D.4. Aligned HC/CR correction equality. Cluster-robust practice rarely uses the unadjusted meat; it applies a small-sample correction from a named HC/CR family, indexed by $a \in \{0, 1, 2, 3\}$, through cluster-level adjustment matrices $\mathbf{A}_j^{(a)}$ acting on the cluster scores or residual contributions before accumulation (Cameron & Miller, 2015; MacKinnon et al., 2023). In the meta-analytic setting these corrections, most prominently the bias-reduced CR2 adjustment, are standard for tests of moderators (Bell & McCaffrey, 2002; Tipton, 2015; Tipton & Pustejovsky, 2015). Writing $\mathbf{A}_j^{(a)} \mathbf{s}_j(\hat{\boldsymbol{\eta}})$ for the adjusted cluster contribution (when a correction is defined at the residual level, this denotes the induced adjusted contribution after the same residual, derivative-design, leverage, and projection conventions have been applied), the corrected conditional meat is

$$\mathbf{M}_\eta^{(a)} = \sum_{j=1}^J \left\{ \mathbf{A}_j^{(a)} \mathbf{s}_j(\hat{\boldsymbol{\eta}}) \right\} \left\{ \mathbf{A}_j^{(a)} \mathbf{s}_j(\hat{\boldsymbol{\eta}}) \right\}^\top, \quad (\text{D.11})$$

and the corrected sandwich is formed with the same named bread under the convention of Section D.1. The display is generic: it defines no new HC/CR family and catalogs no family-specific formula.

Corollary 2 (restated; aligned HC/CR correction equality). Suppose the two interpretations use the same evaluation point, the same likelihood-only cluster scores, the same residuals and derivative design, the same leverage and projection convention, the same adjustment matrices $\mathbf{A}_j^{(a)}$, the same bread convention, and the same cluster partition and ordering inherited from $\mathcal{T}_A$. Then the adjusted cluster contributions coincide cluster by cluster, so the corrected meats coincide,

$$\mathbf{M}_{\eta, \mathbf{M}}^{(a)} = \mathbf{M}_{\eta, \mathbf{S}}^{(a)}, \quad (\text{D.12})$$

and, when the common bread inverse is well-defined under the chosen convention, the corrected sandwiches coincide,

$$\mathbf{V}_{\eta, \mathbf{M}}^{(a)} = \mathbf{V}_{\eta, \mathbf{S}}^{(a)}. \quad (\text{D.13})$$

Proof. By Proposition 2 of Appendix B and the convention of Section D.1, the unadjusted cluster ingredients match cluster by cluster: the evaluation point, the likelihood-only cluster scores, the residuals, and the derivative design. The remaining ingredients (the leverage and projection convention, the adjustment matrices $\mathbf{A}_j^{(a)}$, the bread convention, and the cluster partition and ordering) are stipulated to match. Therefore each adjusted contribution $\mathbf{A}_j^{(a)} \mathbf{s}_j(\hat{\boldsymbol{\eta}})$ is the same vector on the two interpretations, the outer products agree term by term, and summing over the same ordered partition gives (D.12). Assembling the matched corrected meat with the matched bread under the named convention, and taking the inverse where well-defined, or the same estimable-subspace inverse on both interpretations, gives (D.13). $\square$

Corollary 2 is an aligned-definition equality, and it must not be over-read: the two corrected objects agree because every input is matched, and the alignment is the content of the statement. It is not a promise that HC and CR implementations agree, nor that survey and meta-analytic software defaults coincide (those differ in the leverage, residual, adjustment, bread, and degrees-of-freedom choices listed above), and if any ingredient differs the object has changed and the equality is not claimed. A covariance equality alone does not make tests identical: once degrees of freedom are computed differently, the resulting tests differ even when (D.13) holds. The corrected object is still a conditional $\boldsymbol{\eta}$ object, still rank-limited by (D.5); in particular CR2 is an aligned reporting convention the two interpretations share once its ingredients are matched, not the default reported fixed-effect target, which remains the selected $\boldsymbol{\beta}$ -block of the marginal $\boldsymbol{\xi}$ sandwich.

D.5. Relation to Pustejovsky and Chen (2024). The nearest within-meta-analysis precedent for robust-variance correction families is Pustejovsky and Chen (2024), and their analysis already shows that correction equivalence is not automatic: comparing robust variances built from aggregated effect sizes with those from the disaggregated multivariate model, they find equality for the $a = 0, 1, 3$ corrections under their basic equivalence conditions, whereas the bias-reduced $a = 2$ correction of Tipton (2015) requires additional structure (compound-symmetric within-cluster sampling covariance and balanced cluster sizes) and otherwise differs.

Corollary 2 borrows that correction vocabulary but makes a different claim. Pustejovsky and Chen (2024) compare alternative meta-analytic representations of the same data under aggregation or reclassification; Corollary 2 compares two interpretations of the same unaggregated matched Gaussian object, and its conclusion follows internally from shared conditional scores and Hessians together with matched correction ingredients, not from any external equivalence theorem. The matched boundary already imposes a compound-symmetric covariance, the same observation-level design, and the same cluster partition, so it operates in exactly the controlled regime in which an aligned $a = 2$ correction can be the same object on both sides; the alignment of definitions is still required, as the corollary states. The two results are complementary comparators on the same correction families, not the same theorem, and their $a = 2$ equality is not used as proof of Corollary 2.

Appendix E. Deterministic algebra audit

This appendix records the deterministic finite-design audit summarized in Section 5: transcription-and-wiring checks of the algebraic objects in the theorem spine, on a fixed toy design, with expected-failure controls that break one boundary ingredient at a time. The audit is a reproducibility and scope-control record; the pointwise proofs remain in Appendices B–D, and finite evaluations cannot replace them. Two layers are reported because they carry different evidentiary strength: the **conditional/direct layer**, in $\boldsymbol{\eta} = (\boldsymbol{\beta}, \boldsymbol{\theta})$, is a common-evaluator map-wiring check whose matched differences are exact zero by construction once the translation map is wired correctly; the **marginal-vector layer**, in $\boldsymbol{\xi} = (\boldsymbol{\beta}, \tau^2)$, compares two independently implemented likelihood-and-derivative evaluator paths, whose matched differences sit at floating-point scale rather than exact zero.

E.1. Toy design and matched boundary. Both layers use the same fixed design: four clusters with

$$(n_1, n_2, n_3, n_4) = (2, 3, 3, 4), \quad (a_1, a_2, a_3, a_4) = (0.04, 0.09, 0.06, 0.12),$$

and an observation-level design matrix containing a covariate that varies within every cluster, so the audit exercises the unaggregated regime (special condition S1 not invoked; no cluster-mean reduction). Objects are evaluated at three fixed parameter points, labeled **zeroish**, **moderate**, and **off_mode**, crossed with $\rho \in \{0, 0.35\}$, giving six matched-boundary cells per object. These are deterministic evaluation points, not fitted estimates or optimizer outputs, so pointwise identities can be checked without

importing optimization behavior; `off_mode` includes a small τ^2 and supplies the most numerically demanding marginal-curvature cell. Every positive check imposes the matched boundary of Section 3 (same ordered responses, same observation-level $\mathbf{X}_j$, same cluster partition and ordering, fixed matched ρ , matched scales, unit-normalized weights, shared Gaussian kernel), with conditional covariance $\mathbf{V}_j = a_j \mathbf{C}_j(\rho)$ and marginal covariance

$$\Sigma_j(\tau^2) = a_j \mathbf{C}_j(\rho) + \tau^2 \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top. \quad (\text{E.1})$$

The full parameter-dependent determinant $\log |\Sigma_j(\tau^2)|$ is retained. For posterior-bread objects, prior curvature and the evaluation point are matched while the meat remains likelihood-only; for aligned HC/CR objects, the correction family, adjustment ingredients, cluster ordering, and bread convention are held fixed on both interpretations.

E.2. Conditional/direct layer. At each of the six cells, thirteen objects are compared: the cluster log likelihood; total log likelihood; cluster score; total score; likelihood bread; likelihood meat; posterior bread; posterior-bread sandwich; strict cluster constants; and aligned HC/CR adjusted meats for $a = 0, 1, 2, 3$ (the correction index a is distinct from the cluster scale a_j). All checks pass: **82/82 overall**, comprising **78/78 positive checks** and **4/4 negative expected-failure controls**, with every recorded positive difference exactly zero.

The exact zeros must be read according to the architecture that produced them: a single routine evaluates both interpretations after the translation map has aligned their ordered inputs, so equality follows by construction once the map is wired correctly. This layer certifies response, design, and cluster ordering; matched scales and correlations; zero strict cluster constants under the matched normalization; the assembly of score, bread, meat, posterior-bread, and aligned HC/CR objects; and the operation of the boundary-violation switches. It can expose a mistranscribed map, a reordered cluster, an incorrectly retained additive constant, or an inactive control; it is not evidence that two independently derived evaluators agree, because it contains only one evaluator.

E.3. Marginal-vector layer. The marginal layer works in $\boldsymbol{\xi} = (\boldsymbol{\beta}, \tau^2)$ and checks the marginal target of Appendix C. At each fixed audit point $\boldsymbol{\xi}^*$, with likelihood-only marginal cluster scores $\mathbf{s}_j = \nabla_{\boldsymbol{\xi}} \ell_j^{\text{marg}}(\boldsymbol{\xi}^*)$, the meat and sandwich are

$$\mathbf{M}_\xi = \sum_{j=1}^J \mathbf{s}_j \mathbf{s}_j^\top, \quad \mathbf{V}_\xi = \mathbf{H}_\xi^{-1} \mathbf{M}_\xi \mathbf{H}_\xi^{-1}, \quad (\text{E.2})$$

under the named marginal bread convention, together with the posterior-bread analogue (matched prior curvature added to the bread, likelihood-only meat retained) and the selected fixed-effect target

$$\mathbf{V}_\psi^{\text{target}} = \mathbf{R}_\psi \mathbf{V}_\xi^{\text{marg}}[\boldsymbol{\beta}, \boldsymbol{\beta}] \mathbf{R}_\psi^\top, \quad (\text{E.3})$$

checked through a nonidentity selection map sending (β_0, β_1) to $(\beta_0 - \beta_1, \beta_1)$, so that transport is exercised through an actual contrast rather than only the identity. Seven objects are compared per cell: the marginal log likelihood, score, Hessian/bread, cluster-score meat, sandwich, posterior-bread sandwich, and selected block after $\mathbf{R}_\psi$. All checks pass: **48/48 overall**, comprising **42/42 positive checks** and **6/6 negative expected-failure controls**.

The two evaluator paths are genuinely separate. Evaluator A forms the inverse and log determinant by dense linear algebra and differentiates the marginal likelihood analytically; evaluator B forms the inverse through the Woodbury identity and the determinant through the matrix-determinant lemma, then uses complex-step gradients (Martins et al., 2003; Squire & Trapp, 1998) and Richardson-extrapolated Hessians (Richardson, 1911). The paths share neither an evaluation routine nor an inverse/determinant method nor a differentiation method, so their agreement is a stronger finite-design check than the common-evaluator layer, while remaining an audit rather than a proof. The marginal log-likelihood differences are not uniformly exact zero: some cells return exact zero and the largest difference is about 3.6×10^{-15} , at floating-point scale, with all other objects within their prespecified tolerances. A separate analytic-versus-complex-step residual of about 7.8×10^{-9} is a derivative-coding check, not one of the mapped algebraic equality residuals; the complex-step/Richardson construction also avoids the subtractive cancellation that a naive central-difference Hessian encounters in the small- τ^2 `off_mode` cell.

TABLE E.1. Audit-layer summary. “Pass” has a different evidentiary meaning in each row.

Layer	Objects checked	Positive	Controls	Architecture	Evidentiary status
Conditional direct (η)	likelihood, score, bread, meat, posterior bread/sandwich, strict constants, HC/CR meats	78/78	4/4	common evaluator, map wiring	verifies wiring, constants, ordering, controls; not independent-evaluator agreement
Marginal vector (ξ)	marginal likelihood, score, bread, meat, sandwich, posterior-bread sandwich, selected block	42/42	6/6	two independent evaluator paths	stronger finite-design comparison of the reported target; limited to checked objects

E.4. Negative controls. The expected-failure controls are part of the audit design: a positive ledger alone could be satisfied by a comparison that never changes its inputs. Each control removes one boundary ingredient and requires the designated diagnostic to exceed the prespecified 10^{-6} threshold. All ten break as required. The four conditional/direct controls are non-unit weights (altering the cluster contribution and pseudo-score geometry), a mismatched $\mathbf{X}_j$ (altering residual geometry and derivative design), a mismatched ρ (altering the precision matrix), and a scale mismatch (altering the parameter-dependent quadratic, which no cluster-additive constant can absorb). The six marginal controls are scale mismatch, mismatched ρ , mismatched $\mathbf{X}_j$, non-unit weights, a non-Gaussian multivariate- t kernel, and a dropped marginal determinant. The determinant control is the most informative: by the matrix determinant lemma,

$$\log |\Sigma_j(\tau^2)| = \log |\mathbf{V}_j| + \log \left(1 + \tau^2 \mathbf{1}_{n_j}^\top \mathbf{V}_j^{-1} \mathbf{1}_{n_j} \right), \quad (\text{E.4})$$

whose second term depends on τ^2 , so dropping it corrupts the τ^2 score; this is the finite-design counterpart of Caveat 2 in Appendix B. The controls establish that the audited equalities depend operationally on the stated boundary ingredients; they do not characterize behavior throughout the excluded parameter spaces.

The audit supports one narrow conclusion: the checked finite-dimensional objects have been transcribed, transported, and evaluated coherently on the audit design, and the negative controls show that the stated boundary conditions are operationally necessary. It establishes nothing beyond those objects (no operating characteristics, no coverage, no software validation), and six evaluation cells do not replace the pointwise proofs.

Appendix F. Extended empirical results

This appendix expands the empirical illustration of Section 5: the full bridge-identity residual ledger at both working correlations, the complete correlated-effects (CE) fit panels for both syntheses, the external-comparator discipline, the profile-versus-full- ξ comparator exhibit, and the Case A/Case B estimand-existence record. The marginal Godambe sandwich is evaluated at the interior sample score root $\hat{\xi}$, where $\mathbf{H}_\xi$ is positive definite, with the posterior mean as the reported center; the corresponding population pseudo-true root is denoted $\xi^\dagger$ (Appendix C).

Every model reported below is fit in the CE specification with unit weights on the matched homoscedastic boundary at $\rho = 0$,

$$\Sigma_j(\tau^2) = \bar{s}_j^2 \mathbf{I}_{n_j} + \tau^2 \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top,$$

so the study-mean sampling variance $\bar{s}_j^2$ sits on the likelihood diagonal for every observation. Every headline object below, namely the CE fits, the CCC-SD values, and the matched(C) external comparison, which separately refits this same $\rho = 0$ /study-mean working model (Section F.4), uses this working boundary. The $\rho = 0.8$ calculations appear in exactly three labeled roles: the bridge-identity ledger rows certifying the cross-interpretation identity at an alternative fixed working correlation (Table F.2); the CR2($\rho = 0.8$) comparator columns reporting the standard-RVE practitioner contrast under a different working model (Tables F.3 and F.4); and the block-restriction exhibit of Table F.5. None of these is a calibration target, and none is pooled with the $\rho = 0$ headline. Three numerical residuals retain distinct meanings throughout: 0 is the cross-dialect bridge residual; 7.8×10^{-9} is the

derivative-coding residual; and 2.1×10^{-16} and 3.5×10^{-16} are the Clinton and Roberts CCC implementation-target residuals.

F.1. The two illustrative syntheses. The two examples bracket the empirical distinction that organizes the appendix. The **Clinton** reading-performance synthesis (Clinton-Lisell, 2021) contains $J = 20$ studies and $K = 31$ effects; its self- versus experimenter-paced and reading-medium (screen) moderators are constant within every study (0 of 20 studies vary), so Clinton is the degenerate Case-B design in which one-row study aggregation is lossless. The **Roberts** enactment synthesis (Roberts et al., 2022) contains $J = 145$ studies and $K = 443$ effects; its three-level **Comparison_Task** moderator (EPT, IT, and VT) varies within 28 of the 145 studies, spanning 118 effect sizes, so Roberts is the non-degenerate Case-A design in which collapsing a study to one row would replace its task indicators by fractional averages and eliminate the contrast under study (its categorical task moderator could still be represented by a subgrouping construction; see Section F.6).

TABLE F.1. The two illustrative correlated-effects syntheses.

Meta-analysis	J	K	Moderator(s)	Within-study variation	Regime
Clinton (reading performance)	20	31	self- /experimenter-paced; reading medium (screen)	none (0/20)	Case B; aggregation lossless
Roberts (enactment)	145	443	Comparison_Task (EPT/IT/VT)	28/145 studies; 118 effects	Case A; aggregation collapses the contrast

F.2. The bridge identity on both designs. For each design and each fixed $\rho \in \{0, 0.8\}$, the comparison covers the marginal log likelihood, analytic score, observed information, and assembled $\Sigma_j(\tau^2)$ over a grid of $\xi = (\beta, \tau^2)$, with the maximum absolute cross-dialect difference taken jointly over the four objects.

TABLE F.2. Bridge-identity residual ledger. Zero denotes bit-identical agreement under the two interpretations; the $\rho = 0$ rows are the fitted CE working likelihood, the $\rho = 0.8$ rows a representative matched correlation.

Design	ρ	$\max \Delta \ell $	$\max \Delta U $	$\max \Delta I $	$\max \Delta \Sigma_j $	$\max \Delta $
Clinton	0.0	0.00	0.00	0.00	0.00	0.00
Clinton	0.8	0.00	0.00	0.00	0.00	0.00
Roberts	0.0	0.00	0.00	0.00	0.00	0.00
Roberts	0.8	0.00	0.00	0.00	0.00	0.00

Every cell gives $\max |\Delta| = 0$ at machine precision, jointly over the four objects. The $\rho = 0$ rows are the fitted CE likelihood on the matched homoscedastic boundary with common scale $a_j = \bar{s}_j^2$; the $\rho = 0.8$ rows demonstrate the same identity after fixing the same representative correlation on both interpretations. In this precise sense the identity is ρ -free: changing a matched ρ changes the shared covariance, not the equality of its two interpretations, and within-study moderator variation does not disturb it (Proposition 1, Appendix B). A separate deterministic check compared the analytic score and observed information with a complex-step reference; its maximum residual was 7.8×10^{-9} , below the 10^{-5} gate (Martins et al., 2003; Squire & Trapp, 1998). That number checks derivative coding and supplies no additional evidence for the identity. Figure F.1 displays the $\rho = 0.8$ ledger graphically; the ledger, not visual overlap, is the operative record.

F.3. Full CE fits and the selected-block calibration. For a selected fixed-effect block or contrast $\psi = \mathbf{R}_\psi \boldsymbol{\beta}$, the target is $\mathbf{V}_\psi^{\text{target}} = \mathbf{R}_\psi \mathbf{V}_\xi^{\text{marg}}[\boldsymbol{\beta}, \boldsymbol{\beta}] \mathbf{R}_\psi^T$, the selected $\boldsymbol{\beta}$ -block of the marginal $\boldsymbol{\xi}$ -sandwich (Appendix C); the studies-as-clusters vocabulary is that of robust variance estimation and design-based sandwiches (Binder, 1983; Hedges et al., 2010). Tables F.3 and F.4 report the complete panels.

Theorem 1: META and SURVEY interpretations share one marginal likelihood ($\rho = 0.8$)

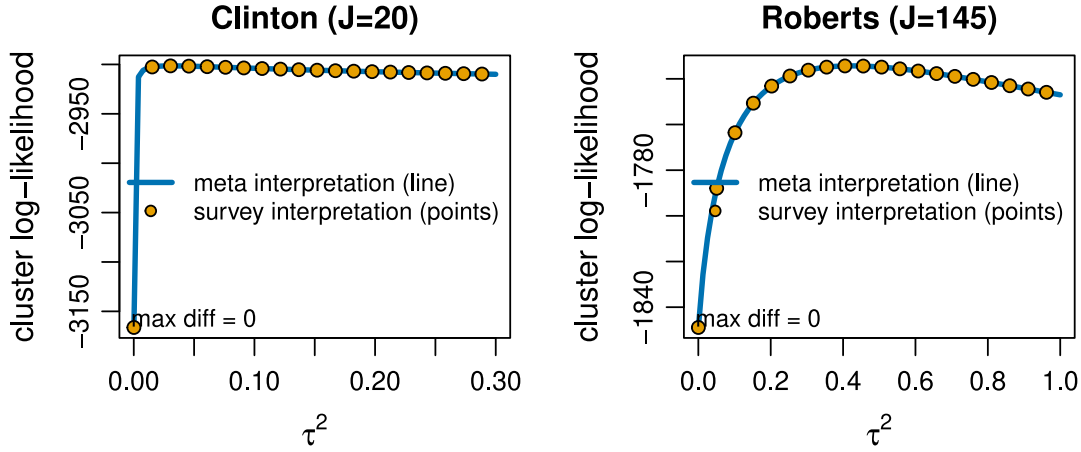


FIGURE F.1. *One marginal likelihood, two interpretations* ($\rho = 0.8$). Each panel plots the marginal cluster Gaussian log likelihood over τ^2 for Clinton ($J = 20$) and Roberts ($J = 145$); the META curve and the SURVEY points coincide because both interpretations evaluate the same function. The displayed residual is the maximum absolute cross-dialect difference over the marginal log likelihood, score, observed information, and $\Sigma_j(\tau^2)$ on the full grid: $\max |\Delta| = 0$ at machine precision here and likewise at $\rho = 0$ (Figure 2). The separately reported complex-step residual (7.8×10^{-9}) checks derivative coding, not the bridge identity.

TABLE F.3. Full Clinton CE panel. The $\text{CR2}(\rho = 0.8)$ column is a labeled observation-level contrast; $\text{matched}(\text{C})$ is the working-model-matched external CR2 comparison.

Coefficient	REML estimate	$\text{CR2}(\rho = 0.8)$ SE (df)	Bayes mean (sd)	CCC-SD (df)	$\text{matched}(\text{C})$ CR2 SE
intercept	-0.550	0.1054 (8.1)	-0.581 (0.1452)	0.1147 (7.1)	0.1162
self-paced	0.355	0.0981 (9.6)	0.396 (0.1289)	0.1137 (11.2)	0.1140
screen	0.102	0.1020 (7.2)	0.074 (0.1387)	0.1053 (9.0)	0.1054

For Clinton, $\hat{\tau} = 0.177$ under REML; the Bayesian posterior mean of τ is 0.213, with median 0.205 and 95% interval [0.126, 0.341]. The calibration preserves the posterior center, including the intercept center -0.581 , while replacing only the selected reported covariance.

TABLE F.4. Full Roberts CE panel. The task moderator varies within study; the $\text{CR2}(\rho = 0.8)$ column remains a labeled different-model contrast, not a parity target.

Coefficient	REML estimate	CR2($\rho =$ 0.8) SE (df)	Bayes mean (sd)	CCC-SD (df)	matched(C) CR2 SE
EPT	0.230	0.1190 (87.0)	0.402 (0.0862)	0.1180 (40.7)	0.1175
IT	0.381	0.1361 (38.5)	0.514 (0.1170)	0.1679 (16.3)	0.1676
VT	1.036	0.0653 (140.7)	1.276 (0.0654)	0.0718 (137.6)	0.0719

For Roberts, $\hat{\tau} = 0.600$ under REML; the Bayesian posterior mean of τ is 0.721, with 95% interval [0.627, 0.828]. The selected-block center is again preserved, including the EPT center 0.402.

The implementation/target check is exact by construction: CCC-SD equals the `bayesRVE` CR2 sandwich standard error to machine precision, $\max |\Delta| = 2.1 \times 10^{-16}$ for Clinton and 3.5×10^{-16} for Roberts. At the reported standard-error layer, CCC-SD is not an approximation to that named sandwich object; it is the selected marginal-sandwich standard error computed from the posterior fit and transported to the reported block without moving its center. These equalities are neither a proof of Corollary 1 nor software validation. One display note carries over to the recoloring figure in the main text (Figure 3): at small J the variance-component prior widens the model-based posterior, so a sandwich ellipse lying inside the posterior ellipse in the Clinton panel is faithful calibration to the narrower named marginal target, not defective shrinkage; in the Roberts panel ($J = 145$) the prior is less influential and the posterior is already tight, and the operation is the same.

F.4. Comparator discipline: `matched(C)`, labeled contrasts, and the ρ -shift decomposition. The phrase “CR2 standard error” is incomplete without its working model. The only like-for-like external comparison is **`matched(C)`**: `clubSandwich`

CR2 (Pustejovsky, 2026) computed after separately refitting the $\rho = 0$ /study-mean working model used by the CE likelihood. Its centers are close to, but not forced to equal, the posterior centers (Clinton intercept $-0.596 \approx -0.581$; Roberts EPT $0.408 \approx 0.402$). Under this matched comparison, CCC-SD and the external CR2 agree to 1.3% for Clinton and 0.4% for Roberts; the remaining discrepancy reflects the expected plug-in and small-sample differences, most visible at $J = 20$.

The CR2($\rho = 0.8$) columns in Tables F.3 and F.4 answer a different question: they are based on heteroscedastic observation-level working models and are labeled contrasts, not parity targets. For Clinton, the column reproduces the published `vcalc` benchmark of Pustejovsky and Chen (2024) in a verified parity calculation; that precedent is not evidence for the bridge or the calibration target. For Roberts, the column is the standard-RVE practitioner contrast, with EPT center approximately 0.23.

The Roberts fit shows the distinction numerically. On the $\rho = 0$ /study-mean model, REML gives $\text{EPT/IT/VT} = 0.408/0.525/1.284$ and $\hat{\tau} = 0.720$, reproducing the Bayesian CE fit to the expected posterior-versus-plug-in tolerance ($0.402/0.514/1.276$; $\hat{\tau} = 0.721$). Approximately 90% of the apparent movement away from the $\rho = 0.8$ REML panel comes from changing ρ from 0.8 to 0; the remainder comes from replacing per-observation sampling variances by the study-mean variances of the CE likelihood. Clinton changes little for the corresponding design reason: many clusters are singletons, the maximum cluster size is 3, and cluster-constant moderators trigger study aggregation, so the within-study off-diagonals have little leverage. In both syntheses the Bayesian fit is faithful to the working model actually fit; the $\rho = 0.8$ comparison differs because it is a different model.

F.5. Profile comparator versus the full- ξ (Godambe/OFS) block. Two covariance comparators to the canonical full- ξ target of Theorem 2 must be kept distinct. The profile β -only comparator selects the fixed-effect block first and uses the bread $\mathbf{H}_{\beta\beta}$; it is the practical approximation. The full-vector Godambe/open-faced-sandwich (OFS) construction sandwiches (β, τ^2) and then reads its β block, using the Schur-complement bread

$$\mathbf{S} = \mathbf{H}_{\beta\beta} - \mathbf{H}_{\beta\tau} \mathbf{H}_{\tau\tau}^{-1} \mathbf{H}_{\tau\beta}^{\top},$$

which additionally propagates τ^2 uncertainty through the information cross-block; equivalently, this β block is the β -block of the full product $\mathbf{H}_{\xi}^{-1} \mathbf{M}_{\xi} \mathbf{H}_{\xi}^{-1}$, not the

reduced form $\mathbf{S}^{-1}\mathbf{M}_{\beta\beta}\mathbf{S}^{-1}$, which would drop the nuisance cross-meat. The two comparators collapse under Cox–Reid orthogonality, $\mathbf{H}_{\beta\tau} = \mathbf{0}$, a condition not met in either design. For this exhibit only, Table F.5 evaluates both constructions on the same matched clusterwise-Gaussian model at the matched-model maximum-likelihood estimate, fixing $\rho = 0.8$ and using CR0 throughout so the comparison isolates the block restriction; it is not an additional CE posterior-fit panel and does not alter the $\rho = 0$ fit-versus-report convention of Tables F.3 and F.4.

TABLE F.5. Profile β -only comparator versus the full- ξ (Go-dambe/OFS) β block at $\rho = 0.8$, CR0 throughout. Forcing $\mathbf{H}_{\beta\tau} = \mathbf{0}$ collapses the difference to machine zero in every cell.

Design	Coefficient	Profile, bread $\mathbf{H}_{\beta\beta}$	OFS block, bread $\mathbf{S}$	Forced $\mathbf{H}_{\beta\tau} = \mathbf{0}$
Clinton	intercept	0.1077	0.1083	0.1077
Clinton	self-paced	0.1086	0.1102	0.1086
Clinton	screen	0.0972	0.0971	0.0972
Roberts	EPT	0.1260	0.1282	0.1260
Roberts	IT	0.1675	0.1692	0.1675
Roberts	VT	0.0689	0.0764	0.0689

For Clinton, the maximum relative difference is 1.49% with $\max |\mathbf{H}_{\beta\tau}| = 166$; for Roberts, 10.89%, attained by the within-study-varying VT cell, with $\max |\mathbf{H}_{\beta\tau}| = 32$. Setting $\mathbf{H}_{\beta\tau} = \mathbf{0}$ collapses the difference to machine zero in every coefficient, isolating orthogonality as the coincidence condition. There is no universal sign to the discrepancy: the two standard errors are diagonals of two distinct sandwiches with different breads, and the positive-semidefinite relation $\mathbf{S} \preceq \mathbf{H}_{\beta\beta}$ imposes no coordinatewise ordering, so an individual coefficient (the Clinton screen cell) can show the OFS standard error marginally below its profile counterpart with no contradiction. On the fitted $\rho = 0$ CE working likelihood, the analogous maximum relative differences are 1.50% for Clinton and 7.63% for Roberts, with $\max |\mathbf{H}_{\beta\tau}| = 175.6$ and 15.5 respectively, which are the values reported in Table 1. Table F.5 uses CR0 so the exhibit concerns only the block restriction; the headline CCC-SD values in Tables F.3 and F.4 are CR2 quantities, and the two versions map onto each other up to the CR2 small-sample factor. The comparison with Wang et al. (2025) is a clarification rather than a correction: their OFS construction begins from an approximate composite

posterior and rescales the full parameter vector, whereas the present analysis begins from the exact shared likelihood, introduces no surrogate, and calibrates a separately named selected-block target; the two constructions answer different questions unless Cox–Reid orthogonality holds.

F.6. Case A versus Case B: the estimand-existence asymmetry. The central distinction between the two syntheses is not a disagreement between two standard errors; it is whether one-row study aggregation can represent the estimand at all. Clinton is Case B: each moderator takes one value per study, aggregation is lossless, and the study-aggregation route of Pustejovsky and Chen (2024) coincides with the studies-as-clusters target, so the degenerate design makes the framework’s additional role nearly invisible. Roberts is Case A for literal collapse: in 28 of 145 studies, covering 118 effect sizes, `Comparison_Task` takes more than one level, and replacing such a study by one row would average EPT, IT, and VT indicators into a fractional task value, destroying the cell-mean contrast. Because `Comparison_Task` is categorical, a Pustejovsky–Chen subgrouping construction could still represent it; Roberts therefore exhibits the failure of literal one-row collapse, while the general estimand-existence claim is strongest for within-study moderators that resist subgrouping altogether. When a within-study moderator cannot be represented by categorical subgrouping, the one-row aggregation estimand has no analogue, while the studies-as-clusters estimating equation and its marginal sandwich remain well defined on the unaggregated observations (subject to the usual rank and estimability conditions), and the bridge identity remains exact at both $\rho = 0$ and $\rho = 0.8$ (Table F.2). The distinction consequently concerns estimand existence rather than the relative size of standard errors, and it is the setting in which the design-based interpretation matters most.

F.7. Scope and provenance. These exhibits realize an identity and a selected-block covariance calibration on one matched unit-weighted clustered Gaussian boundary; they establish no operating characteristics, no coverage, and no extension beyond that boundary (the scope statement of Appendix A governs). The $J = 20$ Clinton discrepancies are left visible: the variance-component prior widens the model-based posterior, and the matched(C) comparison contains a small posterior-versus-plug-in drift; the calibration nevertheless leaves the Bayesian center fixed and changes only the selected reported covariance. The fits, selected-block calibration, and sandwich quantities were produced with the `bayesRVE` package (Lee, 2026; Lee & Tipton, 2026);

that statement records provenance only, and code, the recorded environment, and session information are documented in Appendix G.

Appendix G. Reproducibility, code, and software notes

This appendix records how the results reported in Section 5 and Appendices E–F are regenerated. The complete documentation accompanies the replication package rather than this supplement: the public repository (<https://github.com/joonho112/likelihood-bridge-replication>) provides the ordered analysis scripts and a rendered step-by-step replication guide (<https://joonho112.github.io/likelihood-bridge-replication/>), with an archival DOI to be minted on publication. We record here only what a reader needs without leaving the supplement: the package layout and its two computation tracks, the data inputs and their access route, the recorded computational environment, and the deterministic audit pipelines. The algebra defines and proves the target; the software only realizes it.

G.1. The replication package and its two tracks. The package is a self-contained R project whose analysis code lives in `code/` as nine ordered scripts, numbered 00 through 08 and listed in Table G.1. The likelihood, score, information, and sandwich algebra that several scripts share is factored into two helper files under `code/helpers/` (`cluster_gaussian.R` and `sandwich_marginal.R`), so each algebraic object is defined in exactly one place. The two deterministic audit scripts live under `code/audit/`; `data/` carries a data-access note rather than data files; and `book/` holds the source of the rendered replication guide, which walks through the identity checks, the target construction, the audits, and the empirical exhibits chapter by chapter. The package distributes no raw data and no fitted objects, because every reported number is regenerated from the raw inputs by code, and script outputs are written under `output/`.

Computation separates into a deterministic core and a full empirical track. The default setup installs no Bayesian toolchain and supports the deterministic layer: the two algebra audits, which read no study data, and the real-data likelihood-identity checks, which need only the two raw files. The full track, selected by an environment flag documented in the package, additionally installs `cmdstanr`, `CmdStan`, and the pinned `bayesRVE` package, and it refits both syntheses so that Table 1 and the two empirical figures are regenerated from the raw files rather than read from stored objects.

TABLE G.1. Ordered scripts of the replication package and what each reproduces. Paths are relative to the package’s `code/` directory.

Script	What it reproduces
00_setup.R	Track-aware setup; the full track adds <code>cmdstanr</code> , CmdStan, and the pinned <code>bayesRVE</code> package.
01_prepare_data.R	Reads and recodes the two raw files (Section G.2); verifies the J and K anchors.
02_frequentist_fits.R	REML reference fits; independent <code>clubSandwich</code> CR2 comparator.
03_bayesian_fits.R	Bayesian CE fits at $\rho = 0$ under the fixed seed (Section G.3).
04_likelihood_identity.R	Real-data Theorem 1 ledger: log-likelihood, score, observed information, and assembled $\Sigma_j(\tau^2)$ under both interpretations.
05_marginal_target.R	Definition 2 target at the interior score root; profile and full- ξ standard errors; the center-preserving calibration.
06_profile_vs_ofs.R	Profile-versus-full- ξ exhibits of Appendix F.
07_tables.R	Assembles Table 1.
08_figures.R	The bridge-identity and covariance-calibration figures.
audit/audit_conditional.R	Conditional/direct ledger for Appendix E: 82 checks (78 equality checks, 4 expected-failure controls); base R.
audit/audit_marginal.py	Marginal-vector ledger for Appendix E: 48 checks (42 equality checks, 6 expected-failure controls); Python with NumPy.

The deterministic audit core is the fastest verification because it fits nothing and reads no study data:

LISTING 1. No-data quick start for the audit core, run from the package root.

```
Rscript -e 'source("code/00_setup.R")' \
        -e 'source("code/audit/audit_conditional.R")'
python3 code/audit/audit_marginal.py
```

The two audit scripts are kept outside the empirical workflow because they use finite toy designs and deterministic evaluator checks rather than the two datasets, and the CSV and Markdown ledgers they regenerate, including the expected-failure negative controls, are the authoritative record. They check the theorem algebra on finite designs; they do not replace the pointwise proofs of Appendices B–D, and they license no extension outside the matched boundary.

G.2. Data inputs and access. The empirical illustration uses two third-party data files, and the package does not redistribute them. Both files are distributed in the replication materials that Pustejovsky and Chen (2024) archived on the Open Science Framework (<https://osf.io/9u52e/>); the package’s data-access note records the same instructions together with the column-level expectations of the preparation script. The raw files are to be placed read-only in `data/`; they belong to the original authors and to the Pustejovsky and Chen (2024) materials, which should be cited when the data are used.

The Clinton reading-performance synthesis (Clinton-Lisell, 2021) enters as `readingperformancedatashareable.xlsx`. The preparation script verifies $J = 20$ studies and $K = 31$ effect sizes; both moderators are constant within every study. The Roberts enactment synthesis (Roberts et al., 2022) enters as `Enactment_Meta-Analysis - Behavioural - Data.csv` (the file name contains spaces). The preparation script keeps the rows whose `Effect_Based_On_Hit_Rate` flag is positive, recodes the task labels EPT b and EPT f to EPT and VT d to VT, constructs a within-study effect identifier from row order, and verifies $J = 145$ studies and $K = 443$ effect sizes, with the task moderator varying within 28 of the 145 studies. Because these dimension checks run before any model is fit, a reader whose inputs differ from the analyzed versions receives an immediate error rather than a downstream discrepancy.

G.3. Recorded environment and stochastic reproducibility. Table G.2 records the versions and fixed constants behind the reported runs. This is a recorded snapshot, not a fully locked portable environment (no `renv.lock` or container image is included); the seed and version record support regeneration of the reported stochastic artifacts but do not guarantee draw-by-draw equality across platforms or later releases. The remaining attached packages are recorded in the package documentation.

TABLE G.2. Recorded computational environment and fixed constants for the reported artifacts.

Component or setting	Recorded value	Role
R	4.6.0 (aarch64-apple-darwin)	Statistical computing environment
<code>metafor</code>	5.0-1	Effect-size preparation and frequentist CE fits
<code>clubSandwich</code>	0.7.0	External CR2 comparator under the matched working-model convention
<code>bayesRVE</code>	0.6.0 recorded; 0.7.0 pinned	Produced the reported fits and calibration (0.6.0); the package installs the first tagged public release (0.7.0) and warns on version mismatch
<code>cmdstanr</code> ; CmdStan	0.8.0; 2.38.0	Bayesian sampling backend
Global seed	SEED = 20260614	Fixed seed for the reported stochastic runs
Normalized weights	$\tilde{w} = 1$	Unit-weighted boundary
Fitted CE likelihood	$\rho = 0$	Matched homoscedastic CE likelihood; the reported headline convention
Secondary comparator	$\rho = 0.8$	Labeled comparator in Appendix F, not the canonical target
Likelihood sampling scale	study-mean variance $\bar{s}_j^2$	Common likelihood-diagonal scale on the matched boundary

Both reported Bayesian fits are correlated-effects fits at $\rho = 0$ under the fixed seed. For Clinton, default sampler settings produced maximum $\hat{R} = 1.0020$, minimum bulk effective sample size 1101, minimum tail effective sample size 1259, and zero divergences; the run log contains sporadic proposal-time `normal_lpdf` warnings, and all four chains completed. For Roberts, the recorded settings were 1,500 warmup iterations, 3,000 sampling iterations, and `adapt_delta` = 0.97; the run produced

maximum $\hat{R} = 1.0018$, minimum bulk effective sample size 1291, minimum tail effective sample size 3152, and zero divergences. The reported marginal sandwich is evaluated at the interior sample score root, and the posterior mean serves only as the reported center. These diagnostics document the reported runs; they are not evidence for coverage, draw-by-draw identity, or software correctness.

G.4. Software roles and availability. The `bayesRVE` package fits the Bayesian CE models, computes the named marginal sandwich object, and realizes the selected-block calibration (Lee, 2026; Lee & Tipton, 2026); this statement records implementation provenance only. `clubSandwich` is the external comparator under the stated matched working-model convention (Pustejovsky, 2026); it does not define the target or validate the theorem. No citation in this appendix is assigned a proof role.

The code and its documentation are public. The ordered analysis scripts, the two deterministic audits, and the rendered replication guide live in the replication repository (<https://github.com/joonho112/likelihood-bridge-replication>); the `bayesRVE` package and its documentation are at <https://github.com/joonho112/bayesRVE> and <https://joonho112.github.io/bayesRVE/>. The data-access terms are settled: the raw files are obtained from the archived Pustejovsky and Chen (2024) materials as Section G.2 records, and they are never redistributed with the package.